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BUSINESS PROCESS MANAGEMENT

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CARING PHARMACY

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Introduction

The Order-to-Cash (O2C) is a central operational process within pharmacies. Its performance directly influences service quality, regulatory compliance, employee workload, and overall financial sustainability. Despite its importance, many pharmacies struggle with long waiting times, inefficient resource allocation, and rework caused by external dependencies, such as insurance providers and prescribers. These inefficiencies negatively affect customer satisfaction, operational costs, and the pharmacy's ability to manage increasing demand.

This project aims to analyze, model, simulate, and redesign the O2C process of a pharmacy through a structured Business Process Management (BPM) approach. Using established BPM concepts, including value-added analysis, waste identification, heuristic-based redesign, and hierarchical simulation, the study evaluates the performance of the current AS-IS process and explores alternative TO-BE configurations to improve efficiency and effectiveness.

The research follows the BPM lifecycle, and all process modeling and analysis is conducted using appropriate BPM software and following the BPMN (Business Process Model and Notation) standard. Findings from this project provide actionable insights for pharmacy managers, highlighting opportunities for automation, better resource balancing, and improved customer flow management.

Background

Caring Pharmacy is a well-known pharmacy chain operating across Portugal, with locations in Lisbon, Porto, Faro, Coimbra, Santarém, Torres Vedras, Braga, and Évora. Established in 1998, the company currently employs more than 80 individuals across its various departments. For this project, we will focus on the Lisbon pharmacy as representative of the company's operations. This specific location receives approximately 220 customers daily and is open from 10:00 am to 6:00 pm every day.

The top management at Caring Pharmacy has decided to analyze and upgrade its current business process due to registered customer and employee dissatisfaction, primarily regarding an inefficient and slow process. Issues include the inability of employees to serve all customers during regular work hours, sometimes resulting in staff having to work overtime or customers not being attended to. The goal of this analysis is to overcome these difficulties, increase efficiency, and ultimately improve customer satisfaction. To achieve this, a group of consultants has been hired to model and redesign the actual Order-to-Cash (O2C) process, which is the focus of this project. The scope of the project includes understanding, modeling, analyzing, and redesigning this O2C process.

Business Process Management (BPM) is a structured approach used to analyze, design, execute, monitor and improve organization workflows. Its main goal is to make business processes more efficient, effective, consistent, and adaptable.

Given the pharmacy's high daily customer volume and strict regulatory requirements, the efficiency of the O2C process directly affects operational costs, customer satisfaction, and the ability to deliver timely and safe medication. Any inefficiency in this workflow has an immediate impact on service quality and overall business performance.

BPM Lifecycle

The Business Process Management (BPM) lifecycle provides a structured framework for analyzing and improving organizational processes. It consists of five phases: process identification, discovery, analysis, redesign, and monitoring (Dumas et al., 2018). This project followed these phases to ensure a systematic and evidence-based approach.



Figure 1: BPM Lifecycle

The O2C process was first identified as critical for pharmacy performance. In the discovery phase, the AS-IS workflow was modeled in BPMN using the information provided in the case. The analysis phase combined qualitative techniques (value-added assessment and waste analysis) with quantitative methods (hierarchical simulation) to identify bottlenecks and sources of inefficiency. Based on these insights, the redesign phase applied established redesign heuristics to develop multiple TO-BE scenarios. Lastly, the monitoring perspective guided the evaluation of the redesigned options through operational and financial analysis.

By following the BPM lifecycle, the project ensures methodological rigor and a clear link between analysis and improvement.

Process Identification

The scope of this project is the Order-to-Cash (O2C) process at Caring Pharmacy. This process is defined as a core value chain activity that captures the entire customer-facing flow, beginning with a Customer Request for Medication (the trigger) and concluding with the Delivery and Payment of the fulfilled order. The O2C process is executed collaboratively by the Pharmacy Technician and the Pharmacist. The scope of the current AS-IS model is focused specifically on the Lisbon branch, addressing the high customer volume (around 220 per day) and the observed issues related to inefficiency, slowness, and customer dissatisfaction. This Process qualifies as a business process due to its goal which is generating revenue by meeting customer needs.

Process Discovery (As-Is Model)

Assumptions

To model the "Order-to-Cash" process accurately within the simulation environment, the following operational, structural, and statistical assumptions were established based on the case study data and standard BPM methodology:

Operational Boundaries and Demand

Arrival Pattern: We assume customer arrivals follow a **Poisson Distribution**, reflecting random, independent arrivals throughout the day. Based on the data of **220 customers per day** over an 8-hour shift (10:00 AM – 6:00 PM), the simulation uses a calculated Mean Inter-arrival time of **2.18 minutes**.

Steady State: The simulation assumes the pharmacy operates in a steady state for the duration of the analysis, maximizing the 8-hour window without accounting for specific "warm-up" (opening procedures) or "cool-down" (closing procedures) variances.

Structural Process Logic

Universal Process Path: The model assumes that **100% of customers** follow the main linear process sequence (*Receive* → *Enter Details* → *Check Insurance* → *Fulfill*). This includes the **25% of non-prescription customers** who, despite not requiring an insurance check or clinical review, are currently routed through the same initial bottlenecks as complex patients. This reflects the **lack of an existing triage mechanism**.

Sequential Dependencies: The workflow assumes strict sequential handoffs. The **Technician** must complete the "Receive Client" task before the **Pharmacist** can begin the "Clinical Review" (*Double Check*). There is no parallelism in the AS-IS state as, for example, the Technician does not physically retrieve drugs while the Pharmacist is checking the system.

End-of-Process Payment: It is assumed that payment and invoicing occur strictly at the end of the customer journey, requiring the Technician to be present for the final handover regardless of when the order was physically prepared.

Resource Allocation and Availability

Dedicated Resources: It is assumed that the **4 Technicians** and **3 Pharmacists** are exclusively dedicated to the "Order-to-Cash" process during the simulation run. Auxiliary tasks (such as cleaning, inventory stocking, or shelving) are considered secondary and not modeled as resource-blocking activities in this specific scope.

Skill Differentiation: We assume a strict division of labor based on regulatory and operational roles:

- **Technicians** handle the initial interaction, data entry, physical fulfillment, and payments.
- **Pharmacists** handle clinical reviews, insurance conflicts, and doctor consultations.

Resource Assignment: Tasks involving "Doctor Calls" or "Insurance Replacements" were explicitly assigned to the **Pharmacist** resource in the simulation to accurately capture the constraint on clinical expertise, even if a Technician might initiate the transaction in the system.

Probabilistic Logic

Fixed Probabilities: The decision gateways are modeled using the fixed percentages provided in the project description, assuming these represent a standard daily distribution of work:

- **75%** of customers arrive with a prescription.
- **45%** of prescriptions trigger a Drug Utilization Review (DUR) alarm.
- **80%** of insurance checks require a drug replacement discussion.
- **30%** of the time, the requested medicine is not immediately in stock.

Doctor Availability: We assume that when a doctor must be called, they answer immediately in **55%** of cases. In the remaining 45%, the process encounters significant delays or abandonment, impacting the average cycle time.

Statistical Abstraction of Task Duration

Variability: To account for human variation and complex sub-tasks (e.g., a doctor conversation taking "about 5 minutes" but varying between 3 and 8), all human-centric activities are modeled using a **Truncated Normal Distribution**. This allows the simulation to include realistic outliers (delays) while preventing the generation of mathematically impossible negative time values.

Model Organization

Level 1- Caring Pharmacy Order to Cash

AI-IS Description

The Order-to-Cash (O2C) process at Caring Pharmacy is the core value-chain activity, starting with the customer's request for medication and concluding with the successful delivery and payment of the order. The overall O2C flow involves sequential sub-processes executed by the Pharmacy Technician and the Pharmacist. The process is structured into five key sub-processes: Receiving the Client, Checking Prescription Details, Checking Insurance Coverage, Fulfilling the Order, and finally, Delivering and Handling Payment.

AS-IS Model

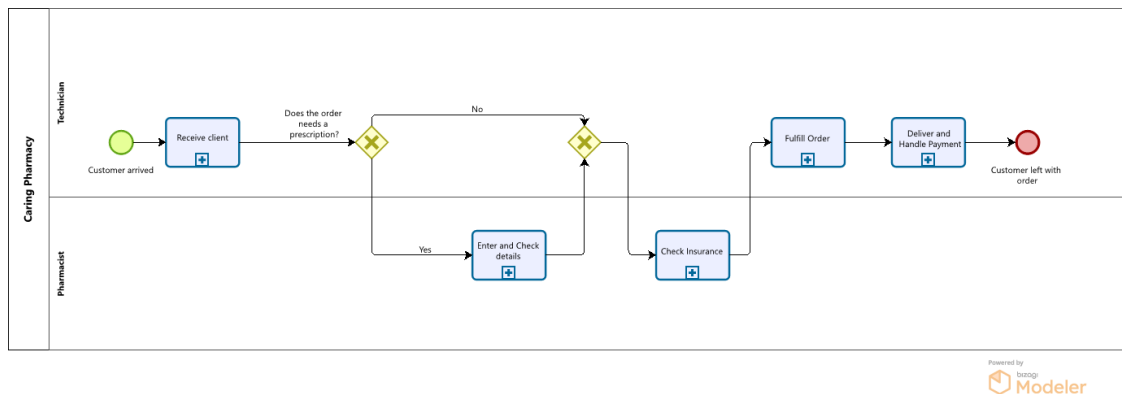


Figure 2: As-Is Order to Cash

Level 2- Receive Client

As-IS Description

This process manages the initial interaction between the customer and the pharmacy technician, starting when the customer arrives at the pharmacy. The technician first checks whether the customer has a medical prescription. If a prescription is provided, the technician receives it and the order is immediately registered, completing the process. If the customer does not have a prescription, the technician records the requested medication and then checks the pharmacy system to verify whether the drug can be dispensed without a prescription. Based on the system's information, the technician determines whether the customer is allowed to pick up the order. If the medication is eligible for pickup, the technician proceeds to register the order; otherwise, the order cannot be fulfilled, and the customer must leave without receiving the medication. This completes the initial intake and validation stage of the pharmacy's order-handling workflow.

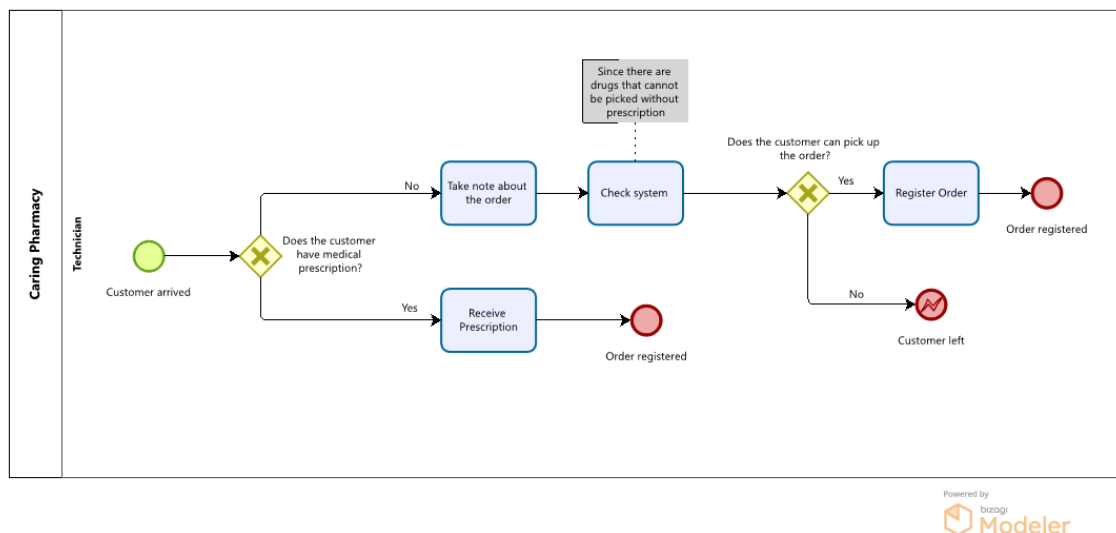


Figure 3: As-Is Receive Client

Level 2- Enter and Check Details

AI-IS Description

The process begins when a customer arrives with a prescription, which the technician enters the system before running an automated Drug Utilization Review (DUR). If no issues are detected, the prescription is approved; however, if the DUR raises any alarms, the pharmacist steps in to review the customer's history and determine whether the patient has a valid reason to receive the medication. If the pharmacist concludes there is no valid reason, they record a note in the system and close the check. If the reason is valid, the pharmacist evaluates whether the doctor may have made an error in the prescription. When an error is suspected, the pharmacist contacts the doctor to clarify or correct the order. If no error is identified, the pharmacist documents the justification in the system. If a corrected prescription is needed and it is electronic, a new prescription is sent; otherwise, the customer leaves the issue handled manually. Throughout the process, steps are recorded in the system to ensure proper verification and documentation.

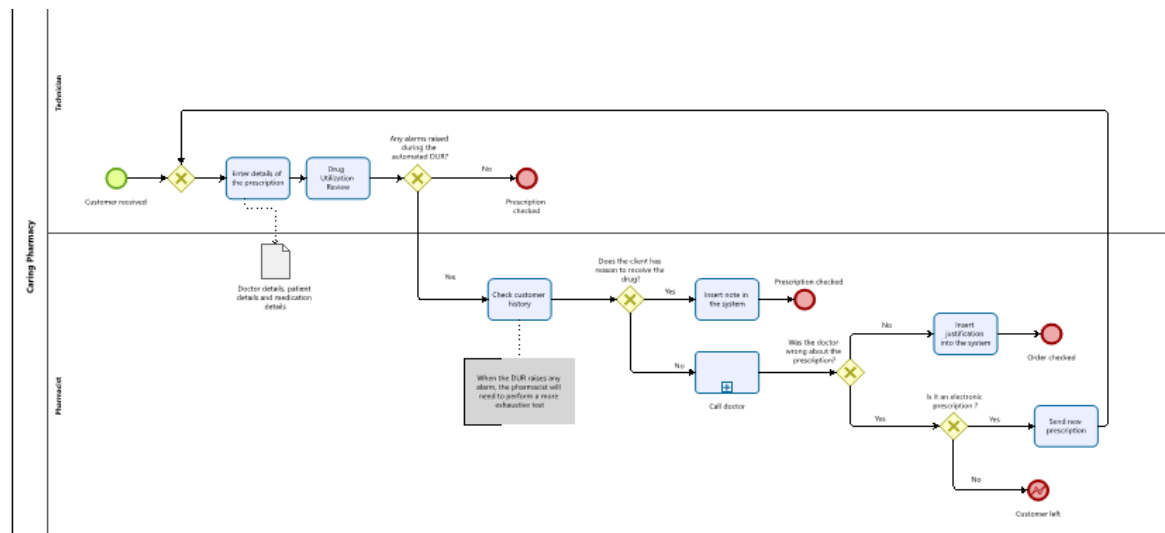


Figure 4: Level 2 As-Is Enter and Check Detail

Level 2- Check Insurance

AI-IS Description

The process begins after the pharmacist has completed checking the order. The first step is to perform an insurance check to determine whether the customer's insurance company will cover part or all the medication cost. If the insurance does cover a portion, the pharmacist uses the company system to calculate the exact amount the insurance will pay, completing the insurance check. If the insurance does **not** cover the cost, the pharmacist discusses possible replacement options with the patient, such as a cheaper or alternative medication. After this discussion, the pharmacist evaluates whether doctor confirmation is necessary for the replacement. If no confirmation is needed, the pharmacist registers the new drug and completes the insurance process. If confirmation is required, the pharmacist contacts the doctor to approve or adjust the prescription. Once the doctor responds, the patient is asked if they want to proceed without insurance coverage. If the customer agrees, the pharmacist registers the order without insurance and finalizes the check. If the customer does not want to proceed, the process ends with the customer leaving without completing the order.

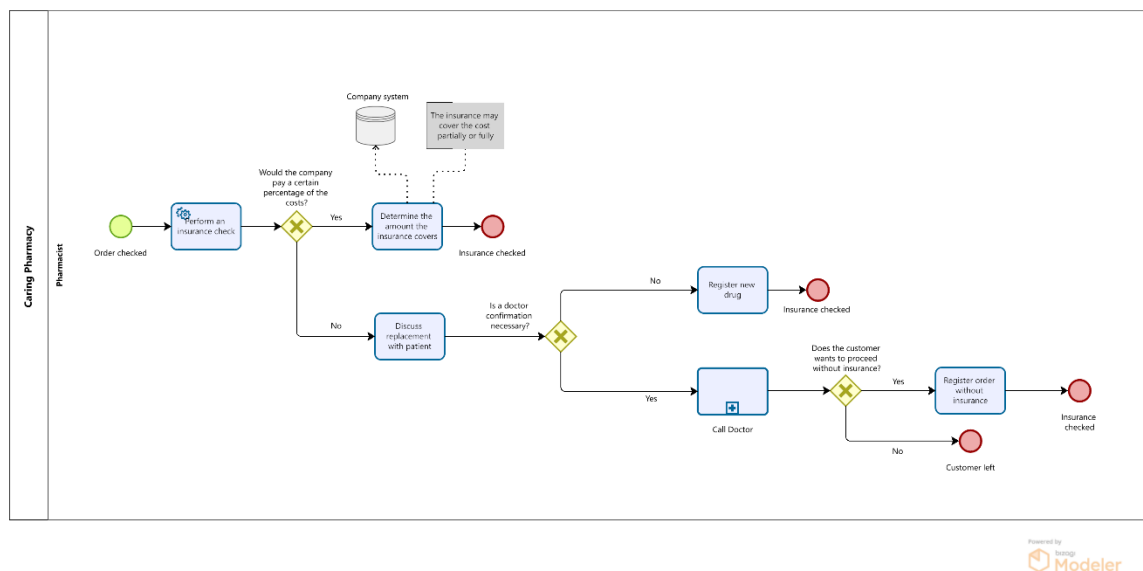


Figure 5: As-Is Check Insurance

Level 2- Fulfill Order

AI-IS Description

The process begins after the insurance has been checked, with the technician verifying whether the required medication is available in stock. If the medicine is in stock, the technician proceeds to grab the drug, after which the order is double-checked. If the drug is confirmed to be correct, the order is packaged and the process is completed. If the drug is not correct, the system triggers a stock recheck to ensure the right product is selected before packaging the order.

If the medicine is not in stock, the technician informs the pharmacist to choose an alternative. The pharmacist first checks whether alternative medication options exist. If no alternatives are available, the patient is informed, and the pharmacy proceeds to order the drug from the supplier, ending the process when the customer leaves. If alternatives do exist, the pharmacist discusses these options with the patient. Depending on the medication, doctor confirmation may be required. If confirmation is needed, the pharmacist will contact the doctor. Once approval is obtained or confirmation is not required, the pharmacist selects the appropriate replacement drug. The selected replacement is then prepared by grabbing the drug, followed by a double-check to ensure accuracy. If the medication is correct, the order is packaged; if not, the process loops back to stock to recheck before packaging. Finally, once everything is validated, the order is fulfilled.

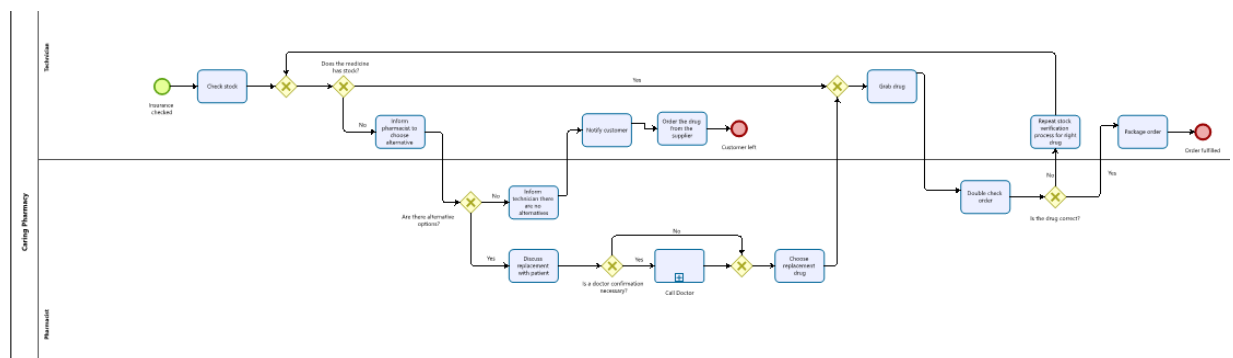


Figure 6: As-is Fulfill Order

Level 2- Deliver and Handle Payment

As-IS Description

The process begins once an order has been fully prepared and is ready for the customer. The technician first receives the payment, confirming the amount and completing the transaction. After payment is processed, the technician delivers the order to the customer. When the order is successfully handed over, the process is concluded.

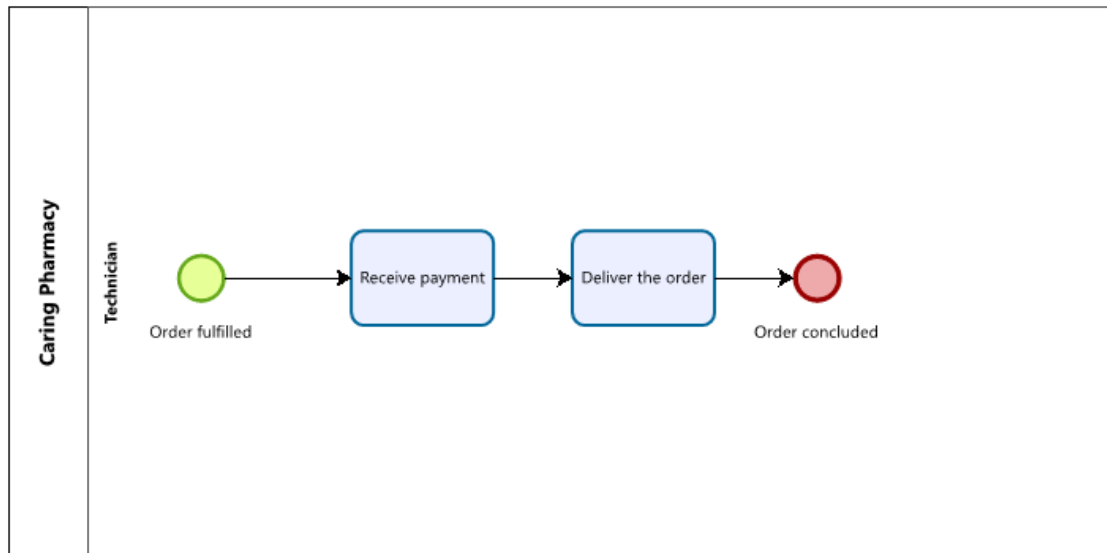


Figure 7: As-is Deliver and Handle Payment

Process Analysis

In business process management (BPM), process analysis is the examination of a business process to understand how it works, how well it performs, and how it can be improved.

In this section of the report, we will present two types of analysis, namely qualitative and quantitative.

Qualitative Process Analysis

Qualitative Process Analysis focuses on understanding a process rather than measuring it numerically; it involves evaluating the characteristics, features, and subjective aspects of the process.

In simple terms, the main goals are to gain deep contextual understanding of the process to identify meaningful improvement opportunities and assess efficacy and efficiency. To better understand the process without relying on hard numerical data, we will use Value-Added Analysis, Waste Analysis, and Root Cause Analysis to uncover insights that cannot be obtained from quantitative metrics alone.

Value-Added Analysis

Value-Added Analysis is a technique used to separate value-adding from non-value adding activities; also evaluates the steps in the process that bring value to the final product or service, however if the step of the process does not bring value, we eliminate it. By removing them, organizations can improve efficiency, overall process performance, and reduce costs.

In Value-Added Analysis, activities are classified into three main types:

1. Value Adding (VA)
 - Activities that directly lead to positive outcomes, contribute to meeting customer needs, and are essential for the final product or service.
2. Business Value Adding (BVA)
 - Activities that are required for legal, policy, or risk reasons, but are needed for the business to operate and do not add value from the customer's point of view.
3. Non- Value Adding (NVA)
 - Activities that do not add any value and are not required for the business.

Firstly, to perform this analysis we will divide each task into steps, then each step will be classified as VA, BVA, or NVA. The tables below will display the classification of the steps in each subprocess from Process Discovery.

Receive Client

Table 1: Receive Client

STEPS	CLASSIFICATION	JUSTIFICATION
Receive Prescription	BVA	Captures information to enable the service; required to proceed but does not transform the service.
Take note of the order	BVA	Administrative capture; necessary to start the work, but no customer value by itself.
Check System	BVA	Legal; protects the firm and patient but does not change the product.
Register Order	BVA	Administrative, records the order so downstream work can happen.

Enter and Prescription Check

Table 2: Enter and Check Details

STEPS	CLASSIFICATION	JUSTIFICATION
Enter details of the prescription	BVA	Data entry enables to automate DUR, although there is no direct customer value, but it is necessary.
Drug Utilization Review	BVA	Mandatory safety review prevents harm, but from a lean view it is a necessary control.
Check customer history	BVA	Clinical compliance steps to ensure safe dispensing; it does not change the product.
Insert note in system	BVA	Documentation required to proceed, essential for internal compliance and professional standards.
Call Doctor	BVA	It is necessary for regulatory and clinical compliance, but it does not directly create value for the customer.
Discuss with the Doctor	BVA	This communication ensures the prescription aligns with clinical guidelines, but it does not directly add value from the customer's perspective.

Insert justification	BVA	Audit requirements to support decisions; however, it does not create value for the customer.
Send new electronic prescription	BVA	It is administrative and compliance work, does not directly create customer value, but it is required by law and regulations.

Insurance Check

Table 3: Insurance Check

STEPS	CLASSIFICATION	JUSTIFICATION
Insurance Check	BVA	Financial eligibility verification, required by external payers and does not bring value to customers.
Calculate the amount of insurance covers	BVA	Necessary for billing and pricing but does not transform the medication.
Discuss replacement with client	BVA	Determines the product the client will receive, directly shapes the outcome, but this step is reworked caused by stock/insurance failure; while it is necessary to close the transaction, it does not add value compared to having the original drug available.
Register new drug	BVA	This does not bring value to the customer; it is only an internal data update.
Call Doctor	BVA	It does not transform the medication, the service, or therapeutic outcome. It only enables the process to continue safely and legally
Discuss with Doctor	BVA	It ensures clinical accuracy and regulatory compliance, but the interaction itself does not create or modify the product that the customer receives.
Register order without insurance	BVA	Administrative recording is needed to process the order but provides no direct customer value.

Fulfill Order

Table 4: Fulfill Order

STEPS	CLASSIFICATION	JUSTIFICATION
Check stock	BVA	Necessary internal control to enable picking; it does not transform the product.
Grab drug	VA	Physically, assembling the order is essential for the customer; however, the walking distance involved in this step represents waste.
Inform pharmacist to choose alternative	BVA	Internal notification; triggers decision but does not change the product.
Discuss with client and doctor	BVA	Co-decides the final medication, directly determines the delivered product, but it only happens due to stockout failure.
Choose Alternative	BVA	Selects the actual drug to be dispensed; transforms the order and it is an internal decision-making forced by lack of inventory
Notify Customer	BVA	Required communication of a constraint; it does not add value to the customer but is necessary.
Order medicine from the supplier	BVA	Triggers replenishment to satisfy the order later, it is necessary but there is no immediate value now.
Double-check order	BVA	This is quality control to make sure the right drug, it does not transform the medication, and it is necessary for safety regulation
Verify correct drug	BVA	Internal availability check is not part of creating the product.
Package Order	VA	It creates the final delivery and directly adds value to the customer.

Delivery and Payment

Table 5: Delivery and payment

STEPS	CLASSIFICATION	JUSTIFICATION
Receive Payment	BVA	Financial settlement, it is necessary for business, however it does not add value for the customer
Deliver Order	VA	Final value realization and then the customer receives the product they purchased.

The value-added analysis shows that most activities in the pharmacy's O2C process are **Business-Value-Adding (BVA)** due to regulatory and safety requirements, while only a small portion directly adds value from the customer's perspective (VA). Overall, the analysis confirms that regulatory constraints limit the removal of steps, making process improvement dependent on redesign heuristics such as **Parallelism**, **Specialization**, and **Automation** rather than task elimination.

Waste Analysis

Waste Analysis consists of the identification and elimination of non-value-added activities that slow down processes, increase costs, and reduce quality. This process aims to identify; reduce all types of waste, also by removing unnecessary activities, organizations can make their processes more efficient, reduce costs, and increase customer satisfaction, in other words improve overall performance.

This analysis presents the seven classical Lean wastes and grouped them into three higher-level categories:

1. Move
 - Transportation: Unnecessary movement of items.
 - Motion: Unnecessary movement or actions of people.
2. Hold
 - Inventory: Holding excess materials or goods.
 - Waiting: Time spent while waiting for the next step in the process.
3. Overdo
 - Defects: Errors or defects requiring rework.
 - Overprocessing: Unnecessary or excessive steps in a process
 - Overproduction: It occurs when more is produced than what is immediately necessary.
 -

The following tables show the diverse waste identified for the entire subprocesses:

Table 6: Waste Analysis: Receive Client

ACTIVITY	MOVE	HOLD	OVERDO	EXPLANATION
Take note about the order			Overprocessing	Repeated administrative work does not change the service.

Table 7: Waste Analysis: Enter and check prescription

ACTIVITY	MOVE	HOLD	OVERDO	JUSTIFICATION
Enter details of the prescription		Waiting	Overprocessing	All prescription details must be manually inserted into the system even though these data are already present in the prescription itself.
Check customer history			Overprocessing	The pharmacist must consult patient history in depth even when it is routine medicine, which extends processing time without improving service outcome.
Insert notes in the system			Overprocessing	Repeated documentation to justify decisions is mandatory but does not change the delivered medication or benefit the customer.
Call Doctor		Waiting	Defects	The process stops completely while waiting for the doctor. Also, there are situations where the doctor fails to answer customer calls; clients frequently leave and return, increasing overall waiting time.
Insert Justification			Overprocessing	Additional documentation that is inserted for audit/compliance reasons, which increases system workload without adding customer value.
Receive new electronic prescription		Waiting		When the doctor sends a new prescription, the customer and pharmacist must wait before the process can resume; waiting is pure delay.

Table 8: Waste Analysis: Insurance check

ACTIVITY	MOVE	HOLD	OVERDO	EXPLANATION
Call Doctor for alternative medication		Waiting	Defects	The process remains blocked until a doctor confirms or rejects a replacement. High doctors' unavailability causes repeated customer delays.

Table 9: Waste Analysis: Fulfill Order

ACTIVITY	MOVE	HOLD	OVERDO	EXPLANATIONS
Inform pharmacist to choose alternative			Defects	This step only happens because the process failed. It is reworking, not doing too much.
Notify customer		Waiting		Notification is needed only because the system lacks real-time inventory synchronization.
Double-check order			Overprocessing	Repetitive verifications only add checking time without enhancing the drug.
Verify correct drug			Overprocessing	Verification ensures accuracy; however, it does not provide value to the customer.

Hold: Waiting (Primary Waste)

This is the direct cause of the **12-minute cycle time** and customer dissatisfaction.

Receive new electronic prescription: When the doctor sends a new prescription, the customer and pharmacist must wait before the process can resume; waiting causes delay.

Call Doctor: The process stops completely while waiting for the doctor (approx. 5 mins).

Manual Data Entry: Customers wait while the Technician manually enters data (0.75 mins)

Discuss replacements with Doctor: A further waiting period while the doctor and pharmacists align on the correct drug, and customers are inactive throughout this delay.

Notify customer (no stock): Notification is needed only because the system lacks real-time inventory synchronization.

Move: Motion

Go get the drug (Technician): This implies the Technician is walking back and forth between the counter and the shelves for every single customer (220 times a day), which is wasted physical effort.

Overdo: Overprocessing

Non-Prescription Customers: The 25% of customers who just want non-prescription items are currently subjected to the same initial intake and queuing as complex patients, resulting in "more work than necessary."

Manual Data Entry: The Technician manually types details that could be scanned or pre-loaded.

Overdo: Defects

No Stock: "In 30% of the cases... the pharmacy does not have the medicines". This forces a massive rework loop (ordering from supplier, notifying customer).

DUR Alarms: 45% of prescriptions raise alarms, requiring the Pharmacist to stop and fix the issue.

Insurance Rejections: 80% of insurance checks fail initially, requiring a "replacement discussion".

Overall, the analysis shows that the most critical wastes in the pharmacy's O2C process are **Waiting** and **Overprocessing**, mainly caused by the heavy workload placed on the Technician, sequential verification steps, and external dependencies such as insurance checks and doctor clarifications. Although some delays stem from unavoidable regulatory requirements (e.g., DUR and pharmacist approval), others, such as repeated data entry, unnecessary movement, and stock-related rework, are avoidable and indicate inefficiencies in the layout. The findings confirm the Technician as the primary bottleneck and highlight the need for redesign heuristics such as **Parallelism**, **Triage**, and **Automation** to reduce waiting, eliminate rework, and improve overall process flow.

Issue Register

An Issue Register is a document or tool that organizations use to record and track issues that arise during a project. The main purpose of the issue register is to record, assess, assign, monitor and manage problems or concerns that may affect the project's productivity.

Table 11: Issue Register

ISSUE	PRIORITY	DESCRIPTION	DATA AND ASSUMPTIONS	QUALITATIVE IMPACT
Slow process and long queues	High	The process is slow and creates long waiting lines	Check prescription for 4 minutes, prepare orders for 2.5 minutes, and 220 customers per day.	Customers wait too long; some leave without being served.
Too much manual work	High	Many tasks require typing and checking by hand	Prescription entry: 0.75 min; 45% DUR alarms need extra work.	Slows the process; increases the errors and they are being less productive
Pharmacists have too many tasks	High	Pharmacists are required for many steps	Pharmacists handle DUR checks, doctor calls, and validations.	Creates long waits, which slows down all customers, and can cause high pressure on pharmacists.

Doctors don't answer	High	Doctors often do not answer phone calls	45% of calls unanswered; customers must return another day.	Can lead to customer frustration; unfinished orders and more work on another day
Insurance Complications	High	Insurance rules can cause many delays and replacements.	In cases where insurance covers part of the cost, 80% require replacement analysis.	The process takes longer; it is more complex, and customers can be unhappy with that.
Medicine out of stock	High	Some medicines are not in stock	70% stock available; 40% of alternatives missing.	Sometimes lost sales or delays, and then the customer must leave without medication
Double-Check medicines	Medium	Pharmacist must re-check what technicians pick	Technician pick: 55 sec; pharmacist check: 35 sec; 10% wrong picks.	Repeat the work, which requires extra time and overall slows down the order preparation
No-Prescription Problems	Medium	Customers without prescriptions may request restricted drugs.	25% of customers have no prescription; 1 min legal check.	Many times, the customers arrive without a prescription, and the drug is not allowed, so the customer cannot get it. Time is wasted on orders that cannot be completed.
Lack of Real – Time Inventory Sync	Medium	System data is not updated in real-time and requires excessive manual input.	A single system handles all steps, but inventory counts are often wrong, and DUR/Insurance checks require manual validation halts.	The system is integrated, but data inaccuracy and manual stopping points (DUR) cause delays, not a lack of connection.

Unpredictable Service Time	Medium	Processing time changes a lot per customer	Depends on DUR alarms, doctor calls, insurance, and stock.	The time needed to serve each customer changes a lot because of DUR alarms, doctor calls, insurance questions and stock issues; some can take 2 minutes, others 10 minutes or more.
Customers must return	High	Many customers cannot finish their order the same day	Happens when the doctor has not answered, wrong prescription, missing stock.	Customers often cannot finish their order because the doctor is not answering the wrong prescription or missing stock. When this happens, they have to leave and come back later. This type of situation increases frustration and reduces trust in the pharmacy.
Order Tracking	High	No system to track the order process.	The process is mostly manual; no ticket or display system	There is no digital queue system or order tracking, and customers do not know how long they must wait, and staff cannot see where delays happen. This leads to confusion, stress and poor organization

The issue register highlights that most operational problems result from the **Technician bottleneck**, complex verification requirements, and external dependencies such as doctors and insurance providers. These issues lead to long waiting times, rework, and inconsistent service levels. Additional challenges include inventory inaccuracies, redundant system interactions, and frequent interruptions for clarifications. Together, these issues confirm the need for a redesigned workflow that redistributes workload, reduces unnecessary tasks, and incorporates automation to improve overall process performance.

Quantitative Analysis

Process Simulation Setup

Methodology: Hierarchical Simulation

To ensure the highest degree of accuracy in our analysis, we adopted a **Hierarchical Simulation** approach. This methodology treats the process as a layered system, distinguishing between the detailed "Child" processes (Level 2 and Level 3) and the high-level "Parent" process (Level 1).

The approach consists of three distinct phases:

Phase 1: The "Child" Analysis (Granular Simulation)

The "Child" processes (e.g., *Fulfill Order*, *Check Insurance*) represent the detailed operational logic. In a standard high-level simulation, these are often oversimplified as single tasks with a fixed duration. However, the project description reveals that these sub-processes contain complex logic, such as:

- **Probabilistic Routing:** e.g., Stock is available 70% of the time, or 30% of the time it is not.
- **Exception Handling:** e.g., Calling a doctor, which introduces significant time variability.
- **Rework Loops:** e.g., Repeating stock verification if the drug is wrong.

Method: We modeled and simulated these Level 2 and Level 3 sub-processes independently. By isolating them, we could observe how these internal probabilities and exceptions affect the total cycle time of that specific activity without the interference of external factors like queuing.

Phase 2: Statistical Abstraction (The "Black Box" Approach)

Once the Child processes were simulated, we analyzed the output to capture the behavior of the activity. We defined each activity using a **Truncated Normal Distribution**, characterized by four key parameters:

- **Mean (Average):** Represents the expected duration of the task derived from the sub-process simulation.
- **Standard Deviation:** Represents the volatility and risk. (e.g., *Fulfill Order* has high volatility due to stockouts).
- **Truncated Limits (Min/Max): This is the critical safeguard.** In a standard Normal Distribution, mathematical probability can sometimes generate negative numbers (impossible in time) or extreme outliers (e.g., a task taking 5 hours instead of 5 minutes). By defining **Minimum** and **Maximum** values based on our sub-simulation results, we ensured the Level 1 model remains physically realistic and stable.

Phase 3: The "Parent" Integration (Systemic Simulation)

The "Parent" process (Level 1) represents the holistic view of Caring Pharmacy. Its primary role is not to calculate task duration, but to manage **Resource Contention** and **Process Flow**.

Method: We utilized the statistical outputs from Phase 2 as the input parameters for Level 1 activities.

- **The Result:** When the Level 1 simulation runs, it accurately predicts how the variability of the children interacts with the limited resources (4 Technicians, 3 Pharmacists).
- **Why this matters:** A simple estimation might suggest the pharmacy has enough capacity. However, our hierarchical approach reveals that when high-variability tasks (like *Check Insurance*) spike in duration, they create "traffic jams" (queues) that block the entire system, leading to the high waiting times observed in our final results.

Choice of Probability Distributions

To represent the variability of the real-world pharmacy accurately, we selected specific probability distributions for different elements of the model. These choices ensure the simulation behaves statistically like the real-life operation.

1. Arrival of Customers: Poisson Distribution

For the Start Event, we selected the **Poisson Distribution** option within Bizagi.

- **Theoretical Justification:** In operations management, customer arrival typically follows a Poisson Process, where customers arrive independently of one another. Mathematically, this implies that the time *between* arrivals follows an Exponential distribution.
- **Bizagi Configuration:** By selecting the "Poisson" distribution in the software, we accurately model this random arrival pattern. We configured the input parameter with a **Mean Inter-Arrival Time of 2.18 minutes** (derived from 480 operating minutes divided by 220 customers). This setting ensures the simulation generates the correct volume of 220 random arrivals over the 8-hour shift.

$$\text{Mean Inter-arrival} = \frac{1}{\lambda} \quad \left| \quad \lambda = \frac{1}{\text{Mean Inter-arrival}}\right.$$

Figure 8: In this equation, λ (**Lambda**) represents the **Arrival Rate** (the number of customers arriving per unit of time). Since our Mean Inter-arrival Time is **2.18 minutes**, our λ is approximately **0.46 customers per minute** ($1 / 2.18$). By inputting the Mean Inter-arrival Time into Bizagi, the software automatically calculates this underlying arrival rate to generate the Poisson process.

2. Activity Durations: Truncated Normal Distribution

For the processing tasks (e.g., *Receive Client*, *Fulfill Order*), we selected the **Truncated Normal Distribution**.

- **Why Normal?** Human-performed tasks naturally follow a "Bell Curve." Most interactions take an "average" amount of time, with some being slightly faster and others slower.
- **Why is Truncated?** During our Hierarchical Analysis, we observed that tasks with high variability (like *Fulfill Order*, which has a high Standard Deviation

due to stockouts) could mathematically generate negative time values in a standard Normal distribution. Since negative time is impossible, and extremely long outliers would distort the results, we used the **Truncated** version.

- **Configuration:** This allowed us to set **Minimum** and **Maximum** boundaries for every task, ensuring the simulation remains physically realistic while still capturing the necessary variability.

3. Machine-Like Tasks: Constant Distribution

For highly standardized tasks, specifically **Deliver and Handle Payment**, we selected the **Constant (Fixed) Distribution**.

- **Theoretical Justification:** This distribution is appropriate for processes with negligible variance, often where the speed is determined by a machine or a rigid protocol rather than human variables.
- **Practical Application:** Our sub-process simulation showed that the payment step had a standard deviation of effectively zero (Avg: 41s). This indicates that the credit card processing and handover interaction is consistent for every customer. Therefore, we modeled this as a fixed duration of **0.68 minutes (41 seconds)** to reflect its stability compared to the highly variable clinical tasks.

4. Decision Gateways: Fixed Probability

For the gateways (e.g., *"Does the order need a prescription?"*), we used **Fixed Probabilities**.

- **Justification:** These are based on historical data provided (e.g., "75% of customers bring a prescription"). These are deterministic ratios rather than time-based distributions, ensuring the token flow will accurately mirror the business's historical volume mix.

Level 2 & 3 Sub-Process Analysis

To be sure about the statistical accuracy of the Level 1 model, we simulated all sub-processes independently. This step validates the probability distributions, task durations, and exceptions accurately, to capture the variability defined in the project description.

Below is the summary of the simulation outputs for all child processes:

SUB-PROCESS	AVG. CYCLE TIME	MAX. CYCLE TIME	KEY VALIDATION INSIGHT
Receive Client	2m 00s	4m 17s	The split between "Receive Prescription" (163 instances) and "Take Note" resulted in a 74% vs 26% ratio, closely matching the 75%/25% requirement.
Check Prescription	1m 55s	6m 18s	The average time includes the "Happy Path" (45s data entry) and the exceptions. The high maximum time validates that the model captures the delays from calling the doctor.
Check Insurance	2m 09s	5m 23s	The "Discuss replacement" task occurred 174 times (79%) , aligning almost perfectly with the 80% probability stated in the case study.
Fulfill Order	2m 48s	10m 41s	The average is extremely close to the project description's 2.5 minutes . The Max Time of 10m 41s captures the complex stockout scenarios involving supplier orders.
Payment	0m 41s	0m 41s	The process is constant with zero deviation, confirming the standard procedure of receiving payment (35s) and handing over the order (6s).
Call Doctor (Level 3)	2m 40s	25m 19s	Validates the high variability of doctor availability. The specific task "Discuss issue with doctor" averaged 4m 12s , aligning with the "approx. 5 minutes" rule.

Table 12: Sub-Process Simulation Results Summary

Detailed Analysis

Receive Client: This process handles the initial routing of customers. The simulation confirmed that most instances (163 out of 220) followed the prescription path (1m 30s avg), while the non-prescription path remained faster (30s avg). The resulting weighted average of **2 minutes** serves as the baseline input for the Level 1 start phase.

Enter and Check Details: This process is highly sensitive to exceptions. While basic data entry is fast, the simulation reveals an average cycle time of **1m 55s**.

Level 3 Validation: We explicitly simulated the "Call Doctor" sub-process to understand the delays. The results showed that while the average call interaction is **2m 40s**, the task "Discuss issue with doctor" averaged **4m 12s**. This confirms our model accounts for the significant time investment required when a pharmacist must clarify a prescription.

Insurance Check: This simulation confirmed that insurance issues are the norm, not the exception. With a **79%** occurrence rate for replacement discussions (174 instances), the average cycle time settles at **2m 09s**. This validates the decision to assign this task to the Pharmacist in the Level 1 model, as it represents a consistent, time-consuming workload.

Fulfill Order: This physical process involves stock checking and retrieval. The simulation output of **2m 48s** aligns with the project description (approx. 2.5 mins). Significantly, the "Are there alternative options?" gateway was triggered **61 times**, indicating a realistic stockout rate where technicians must spend extra time finding solutions, pushing the maximum cycle time to over 10 minutes for those unlucky customers.

Payment: As expected, this process showed no variability (Standard Deviation = 0). The total time of **41 seconds** combines the fixed times for receiving payment and delivering the order, serving as a constant verification step at the end of the customer journey.

Derivation of Level 1 Input Parameters

Following the validation of the sub-processes, we combined the statistical outputs to configure the Level 1 Parent simulation. This step certifies the main model inherits the precise variability observed at the detailed levels.

Table 13 below summarizes how the specific average durations and maximum boundaries observed in the child processes were directly mapped to the configuration of the parent activities.

LEVEL ACTIVITY	1	SOURCE SUB-PROCESS	DERIVED MEAN	DERIVED STD. DEV*	MIN MAX TIME**	DISTRIBUTION SELECTED
Receive Client		<i>Receive Client Analysis</i>	2m 00s	0m 30s	0m 30s / 4m 17s	Truncated Normal
Enter & Check Details		<i>Check Prescription</i>	1m 55s	1m 25s	0m 38s / 6m 18s	Truncated Normal
Check Insurance		<i>Insurance Check Analysis</i>	2m 05s	0m 50s	0m 24s / 5m 23s	Truncated Normal
Fulfill Order		<i>Fulfill Order Analysis</i>	2m 48s	1m 35s	1m 09s / 10m 41s	Truncated Normal
Deliver & Payment		<i>Payment Analysis</i>	0m 41s	0m 00s	0m 41s / 0m 41s	Constant

Table 13: Level 1 Input Configuration

*Standard Deviations are estimated based on the spread between the Average and Maximum times observed in the sub-process results. **Min/Max values correspond to the absolute minimum and maximum cycle times recorded during the 220-instance sub-process simulations.

Note: The high Maximum Time for 'Fulfill Order' (10m 41s) captures the specific exception path where stock is missing, and the Technician must search for alternatives or contact a supplier, as modeled in the sub-process analysis.

Level 1 Simulation Analysis (Global Results)

The final simulation of the Level 1 "Order-to-Cash" process integrates the variability derived from the sub-process analysis to confirm the "As-Is" performance.

Methodology: The Collapsed Process Abstraction

In the final Level 1 model, all subprocesses (e.g., *Fulfill Order*) were kept collapsed and treated as statistical "Black Boxes." This decision was crucial for the Hierarchical Simulation methodology:

- **Efficiency:** It prevents the simulation from processing the detailed logic (Level 2/3) multiple times when running the aggregate flow, significantly speeding up the calculation.
- **Abstraction:** The Level 1 simulation is designed to study resource contention and queuing (the external factors), not task-level details. By using the Truncated Normal Distribution for each activity, the model accurately reflects the overall time and variability of the "Child" processes while maintaining a clear, high-level overview.

Final Throughput and Cycle Time

The simulation successfully processed **100% throughput** (220 instances), confirming the pharmacy has sufficient absolute staff capacity to handle the current daily order volume.

The **final average time** for a customer to complete the process is **12 minutes and 0 seconds**. This efficiency analysis shows a significant **inefficiency gap**, meaning, while the total time spent on value-added work is approximately 10 minutes, this **2-minute** difference confirms customers spend roughly **17% of their total cycle time waiting** due to queuing and resource contention.

NAME	AVG. CYCLE TIME	AVG. WAIT TIME
Caring Pharmacy (Process)	12m 0s	N/A
Receive client	2m 35s	0m 26s
Enter and Check details	2m 21s	0m 7s
Fulfill Order	4m 18s	0m 31s
Deliver and Handle Payment	1m 12s	0m 33s
Check Insurance	2m 20s	0m 6s

Table 14: Final Simulation Time Results (Level 1 Process)

Bottleneck Identification and Verdict

The analysis confirms that the system is stable but exhibits a clear resource constraint. The Technician is confirmed as the **bottleneck resource**, in other words, the single element running at the highest utilization and, therefore, dictating the system's throughput.

RESOURCE	UTILIZATION	STATUS
Technician	73.76%	Primary Bottleneck
Pharmacist	58.87%	Slack Capacity

Table 15: Bottleneck identification

Final Verdict on the AS-IS Model

The analysis confirms that the system is stable but exhibits a clear resource constraint, which is the Technician. The high utilization of the **Technician role (73.76%)**, who is responsible for three of the five Level 1 activities, is the definitive cause of the small but persistent queues at the beginning (*Receive Client*) and end (*Fulfill Order, Deliver/Payment*). This validates customer dissatisfaction and establishes the primary target for the **To-Be** redesign: **offloading the Technician's transactional workload**.

What-If Analysis (Scenarios for Redesign)

To address the identified bottleneck (Technician saturation) and the inefficiency gap, we developed and simulated three cumulative scenarios. These scenarios apply specific **Process Redesign Heuristics** to test the potential performance of the To-Be state before implementation. All scenario results presented below were obtained by modifying the parameters of the validated Level 1 simulation model

Scenario 1

Logic Optimization (The "Green Lane")

Heuristic Applied: Triage. The goal was to stop forcing non-prescription customers (25%) through unnecessary clinical checks.

Configuration: We modified the process logic to route 25% of the flow to bypass the "Enter Details" and "Check Insurance" tasks, proceeding directly to fulfillment. **The simulation output revealed** that while Pharmacist utilization dropped to 46.73%, the overall cycle time remained stagnant at 12m 04s, proving the bottleneck had not moved.

Simulation Results:

- **Pharmacist Utilization:** Dropped significantly from **60.28%** (As-Is) to **46.73%**. This confirms that the logic fixes successfully reduced the clinical workload.
- **Technician Utilization:** Remained high at **74.86%**.
- **Avg. Cycle Time: 12m 04s.**

Conclusion: While this scenario successfully optimized the Pharmacist's role, it had **a negligible impact on the customer cycle time**. This quantitatively proves that the Technician is the *hard constraint* of the system. Thus, freeing up the Pharmacist does not speed up the process if the Technician is still busy handling the start and end of the flow.

Scenario 2

Front-End Automation (Self-Service Kiosk)

Heuristic Applied: Automation (Front-End) and Re-sequencing.

Configuration: Building on Scenario 1, we introduced a Self-Service Kiosk to handle data entry. To quantify the impact of the Kiosk, we updated the 'Receive Client' task duration in the simulation model to a Mean of 0.75 minutes (reflecting assisted self-service). **The resulting simulation data showed** a direct correlation: Technician utilization fell to 55.74%, and the average cycle time dropped to 9m 50s.

Simulation Results:

- **Technician Utilization:** Dropped from **74.86% to 55.74%**. This is a massive improvement, finally creating necessary "slack" time for the bottleneck resource.
- **Avg. Cycle Time:** Reduced to **9m 50s**.

Conclusion: By breaking the bottleneck at the start of the process, we eliminated the initial queue. Customers now move through the reception stage efficiently, saving approximately 2 minutes per person compared to the As-Is state.

Scenario 3

Fulfillment Automation (Green Lane + Kiosk + Robot)

Heuristic Applied: Automation (Core Process) and Parallelism.

Configuration: We configured the final simulation with the cumulative changes: the Green Lane logic, the Kiosk times, and a reduced 'Fulfill Order' duration of 1.0 minutes to represent the Robot's speed. **The final simulation results confirmed** a dramatic improvement, yielding a validated Average Cycle Time of **5m 44s** and a Technician utilization of **22.04%**

Simulation Results:

- **Technician Utilization:** Dropped to **22.04%**.
- **Pharmacist Utilization:** Stabilized at **52.60%**. This is the perfect outcome of Green Lane, ensuring the Pharmacist is not overworked despite the faster throughput.
- **Avg. Cycle Time:** Reduced to **5m 44s**.

Conclusion: This scenario represents the optimal "To-Be" state. The process is now incredibly fast, with the average customer leaving in under 6 minutes. The low Technician utilization (22%) indicates that the pharmacy has effectively **tripled its capacity** and could handle significant growth without creating new queues.

Summary of Improvements

The table below summarizes the cumulative impact of our redesign scenarios against the baseline.

METRIC	AS-IS BASELINE	S1 (LOGIC FIX)	S2 (+ KIOSK)	S3 (+ ROBOT + GREEN LANE)	TOTAL IMPROVEMENT
Avg. Cycle Time	12m 00s	12m 04s	9m 50s	5m 44s	52% Faster
Technician Util.	73.76%	74.86%	55.74%	22.04%	Load Slashed
Pharmacist Util.	58.87%	52.60%	63.55%	52.60%	Perfectly Balanced

Table 16: Comparison of Simulation Scenarios.

Note: S3 represents the final performance expected from the To-Be Model, incorporating the Logic (S1), Kiosk (S2), and Robot (S3).

Conclusion and Selection of the To-Be Scenario

The What-If analysis provides definitive quantitative evidence that neither logical changes alone (Scenario 1) nor partial automation (Scenario 2) are sufficient to fully optimize the Caring Pharmacy.

Selection of the Optimal Scenario

The results demonstrate that the "As-Is" inefficiencies originate from two distinct root causes, that require a combined, synergistic solution. While the logical optimization of the "Green Lane" (Scenario 1) was essential to stabilize the **Pharmacist** utilization at a healthy **52.60%**, it failed to speed up the process. It was only through the introduction of physical automation (Kiosk and Robot) that we could break the **Technician's** constraints, drastically reducing their utilization to **22.04%**.

Therefore, we have selected **Scenario 3 (Cumulative)** as the basis for the formal **To-Be** design. This scenario obtains the optimal balance of performance and scalability, reducing the Average Cycle Time from 12 minutes to **5 minutes and 44 seconds**, a 52% improvement that virtually eliminates customer waiting time. Furthermore, by lowering Technician utilization to below 23%, the pharmacy gains the capacity to handle significant future demand surges without degrading service quality.

Having validated the performance parameters of this scenario, the following section will detail the structural changes and specific heuristics required to implement this new workflow in reality.

Process Redesign

Process Redesign is about making crucial improvements in how a business operates. In this phase, the objective is to achieve meaningful gains in critical performance measures, specifically cost, quality, service, and speed. It focuses on delivering value to customers, which often requires using technology to streamline processes and align teams working across different parts of the organization.

In this step, we aim to eliminate waste and redundancies, reduce errors and defects, become more customer-centric, deliver products/services faster, and finally, use resources more effectively to remain competitive.

There are generally three types of approaches to Process Redesign:

- Explorative Redesign (Transformational).
- Exploitative Redesign (Transactional).
- Reference Model-based Redesign.

Selected Approach: For this project, we adopted an **Exploitative Redesign (Transactional)** approach. We focused on analyzing the existing "Order-to-Cash" process to identify specific operational weaknesses (such as the Technician bottleneck) and applied solutions to resolve them without altering the fundamental nature of the business.

Redesign Heuristics

To execute this exploitative approach, we utilized **Redesign Heuristics**. These consist of a set of proven, experience-based rules of thumb that guide the transition from an existing AS-IS process to a new TO-BE process design. In process redesign, they are used to systematically address problems and weaknesses identified in the current process to achieve improvements in core performance dimensions, namely, cost, time, quality, and flexibility.

In other words, a heuristic is a pragmatic rule that guides problem-solving, which, while not guaranteeing the absolute optimal solution, provides a quick and highly effective solution based on past successes.

Consequently, we used Heuristic Redesign Methodology to systematically solve the identified bottlenecks, leveraging radical, technology-based solutions to achieve necessary improvements in speed and efficiency.

Assumptions for the To-Be Model

The feasibility of the proposed "To-Be" model and the validity of the simulation results rely on the following operational, technological, and behavioral assumptions:

Structural Logic

Safety of the "Green Lane": We assume that the **25% of customers** purchasing non-prescription items require absolutely no clinical intervention from a Pharmacist. This validates the logic of the "Green Lane" that routes these tokens directly to the fulfillment stage.

Digital Clinical Review: We assume that the Pharmacist can perform a valid "Clinical Safety Check" using only the digital data provided by the Kiosk (scanned prescription, patient history, and DUR alerts) without needing to physically inspect the drug package first. This assumption enables the **Parallelism** between the Pharmacist (Safety Check) and the Robot (Retrieval).

Technological Capabilities & Integration

Real-Time Inventory Accuracy: The elimination of the manual "Check Stock" and "Search for Alternatives" tasks is based on the assumption that the Automated Dispensing System (ADS) maintains **>99% inventory accuracy** and is fully integrated with the Kiosk. If the Kiosk accepts an order, the stock is assumed to be physically available for the Robot to retrieve.

Automation Reliability: The simulation parameters assume a high reliability for the hardware:

- The **Kiosk** is assumed to process standard data entry in **45 seconds** with a **95% success rate** (only 5% of cases require Technician intervention).
- The **Robot** is assumed to retrieve and dispense standard items in **1 minute** consistently (Standard Deviation = 0.06 min).

Resource Roles and Strategy

Role Transformation: It is assumed that the **Technician's** primary role shifts from "Transactional Execution" (typing, searching, counting) to "Supervision and Logistics" (exception handling, final packaging).

Staff Retention: The model assumes the retention of all **4 Technicians** despite the low utilization (13%). This is based on the strategic assumption that this "slack capacity" is required to absorb the projected demand growth and provide operational resilience against absenteeism.

Customer Adoption and Payment

Self-Service Adoption: We assume a high willingness among customers to use the Self-Service Kiosk. The model accounts for a minority of customers (modeled as exceptions) who may require manual assistance.

Digital Payment: It is assumed that **100% of standard payments** are processed at the Kiosk or via mobile app prior to fulfillment. This justifies the elimination of the manual "Receive Payment" task at the handover counter.

Financial and Market Behavior

Demand Elasticity: The financial analysis assumes that customer demand is elastic regarding service speed. We assume that reducing the cycle time from 12 minutes to <6 minutes will directly result in a **10% increase in daily order volume** due to reduced churn (customers leaving lines) and improved market reputation.

Presentation of Solutions and Heuristics

To transition from the "As-Is" state to the optimized "To-Be" model, we implemented three core solutions. Each solution applies specific redesign heuristics and was evaluated against the **Devil's Quadrangle** dimensions to assess the strategic trade-offs.

Task-Level Heuristics

1. Elimination: We applied this heuristic by completely removing the manual "**Print Invoice**" and "**Receive Payment**" tasks from the end of the process. By integrating digital invoicing (via email/app) and automated payment at the Kiosk, these non-value-added steps were eliminated, replacing a **41-second manual task with a rapid 10-second handover**, while preventing the Technician from being detained at the final stage.

2. Decomposition: We decomposed the "**Fulfill Order**" sub-process, which was previously a single, long task performed by the Technician. In the To-Be model, we split it into two distinct components: "**Drug Retrieval**" (assigned to the Robot) and "**Packaging**" (assigned to the Technician). This separation allowed us to automate the physically demanding part while maintaining the necessary human touch for the final check.

3. Triage: To optimize flow entry, we implemented a triage gateway at the very start of the process (The "**Green Lane**"). Instead of treating all customers as complex cases, the process now differentiates between "Prescription" and "Non-Prescription" arrivals immediately. This allows **25% of customers** to be routed directly to fulfillment, avoiding the unnecessary "Enter Details" and "Check Insurance" tasks.

Flow-Level Heuristics

4. Re-sequencing: We re-sequenced the "**Check System/Data Entry**" validation to occur at the very beginning of the process via the Self-Service Kiosk. In the As-Is model, errors were often caught late; moving validation to the entry point ensures exceptions are caught immediately, therefore preventing invalid orders from consuming downstream resources.

5. Parallelism Enhancement: We restructured the sequential workflow between the **Pharmacist** and the **Dispensing Robot**. Previously, the Technician waited for the Pharmacist to finish the "Clinical Review" (*Double Check*) before searching for the drug. In the To-Be model, the Robot retrieves the drug *simultaneously* while the Pharmacist performs the clinical check, significantly reducing total cycle time through task overlapping.

Process-Level Heuristics

6. Specialization and Standardization: We standardized the order intake process using the **Kiosk**, which ensures every order follows a consistent digital format and reduces errors. This standardization allows human roles to specialize: the **Pharmacist** focuses purely on high-value clinical reviews, while the **Technician** specializes in customer interaction and logistics rather than data entry.

7. Resource Optimization: Guided by the simulation results, we strategically shifted workload away from the **Technician** (the bottleneck, at 76% utilization) to the **System**. This allowed us to balance the Pharmacist's utilization (stabilized at 52%), ensuring neither human role is overloaded.

8. Communication Optimization: We optimized information flow by replacing paper trails with **Digital Invoices** and **System Alerts**. This ensures instant communication between the Front Desk (Kiosk), Back Office (Robot), and Customer (App), eliminating the delay inherent in printing and physically moving papers.

9. Automation: Finally, automation served as the enabling heuristic for the entire redesign. We deployed technology to handle the repetitive aspects of the process: the **Self-Service Kiosk** for data entry and the **Automated Dispensing System (Robot)** for physical retrieval. This provided the raw speed and consistency required for the other heuristics to function effectively.

Evaluation of Redesign Impact (The Devil's Quadrangle)

Following the application of the redesign heuristics, we evaluated the holistic impact of the proposed "To-Be" model against the four dimensions of the **Devil's Quadrangle**: Time, Cost, Quality, and Flexibility.

This evaluation highlights the strategic trade-offs accepted to achieve the project goals.

DIMENSION	IMPACT DIRECTION	EVALUATION & JUSTIFICATION
Time	↓ Decrease (Positive)	Primary Goal Achieved. The combination of <i>Automation</i> , <i>Parallelism</i> , and <i>Triage</i> reduced the average cycle time by 54% (from 12m to 5.5m). The "Inefficiency Gap" (waiting time) was effectively eliminated.
Cost	↑ Increase (Negative)	Accepted Trade-off. Implementing the <i>Automation</i> heuristic requires a significant initial CAPEX of €96,500 (Robot + Kiosk). However, this short-term cost increase is justified by the rapid ROI (1.6 years) driven by capacity growth.
Quality	↑ Increase (Positive)	Service & Safety Enhanced. <i>Standardization</i> via the Kiosk reduces data entry errors. <i>Specialization</i> allows the Pharmacist to focus purely on clinical safety without distraction. The Robot eliminates picking errors, directly improving patient safety.
Flexibility	↑ Increase (Positive)	Scalability Unlocked. The <i>Resource Optimization</i> heuristic created massive "slack" capacity (Technician utilization <13%). The system is now highly elastic, capable of handling a 300% demand surge without degrading service, unlike the rigid "As-Is" process.

Table 17: Evaluation of Redesign Impact

Conclusion on Trade-offs: The redesign represents a strategic decision to accept a **financial investment (Cost)** in exchange for transformative and permanent improvements in **Time, Quality, and Flexibility**.

TO-BE Process Models

The following BPMN diagrams illustrate the structural implementation of the redesign solutions. These models visualize how the theoretical heuristics, mainly Triage, Automation, and Parallelism, were translated into specific process flows to resolve the Technician bottleneck.

Level 1 Process Map: The "Green Lane" Triage

This diagram highlights the fundamental structural change: the introduction of a **Triage Gateway** immediately after the start event. By applying the **Triage heuristic**, we created a distinct "Green Lane" for non-prescription customers (25%) that physically bypasses the Pharmacist Lane, following directly to the parallel fulfillment stage. Additionally, the resource lanes have been restructured, meaning, the Technician Lane is simplified to reflect their reduced role, while new "System" lanes (Kiosk, Robot) are introduced to visualize the integration of automation.

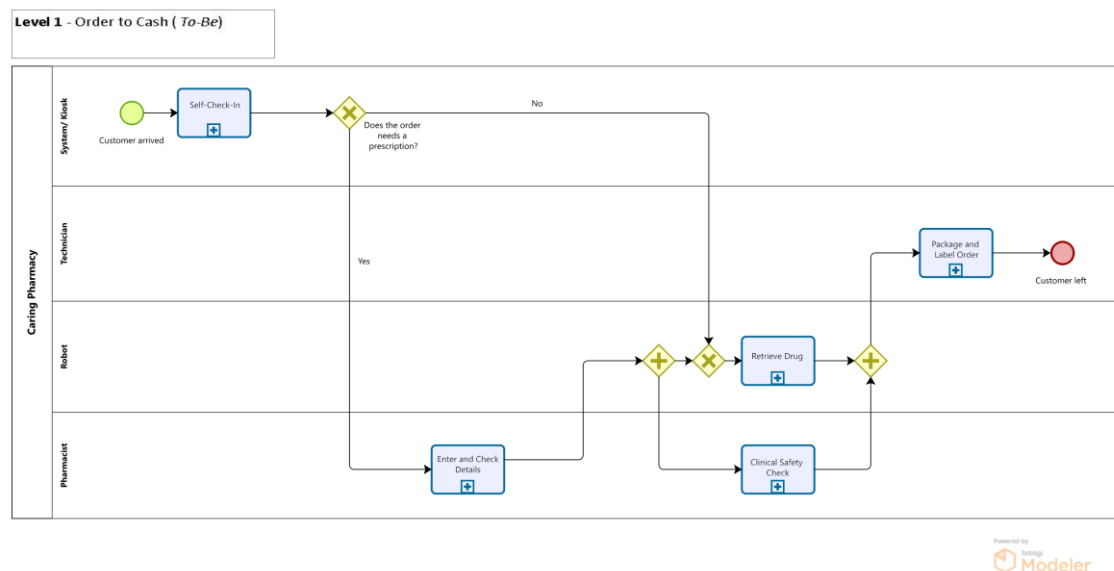
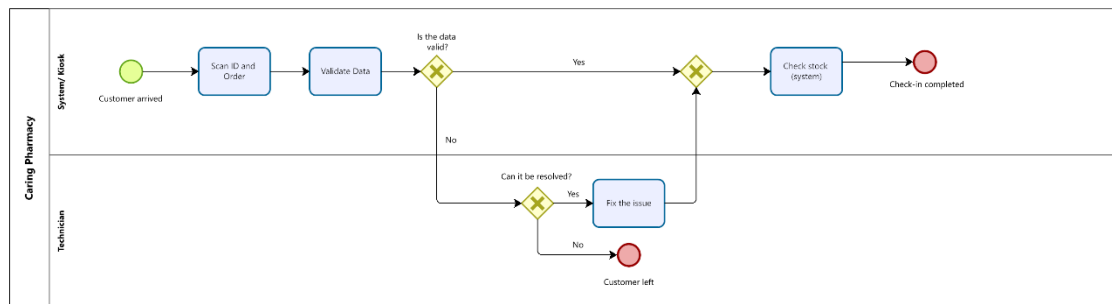


Figure 9: Level 1 To-Be Diagram

Sub-Process: "Self-Check-In" (Kiosk Automation)

Applying the **Automation** and **Re-sequencing heuristics**, this diagram details the transformation of the front desk workflow. The manual "Receive Client" task previously performed by the Technician has been replaced by a "**Customer Self Check-in**" workflow situated in a new "**System / Kiosk**" lane. This system handles data validation immediately upon arrival. Consequently, the Technician is removed from the critical path and reserved solely for a supervisory role ("Resolve Exception"), effectively eliminating the initial queue.

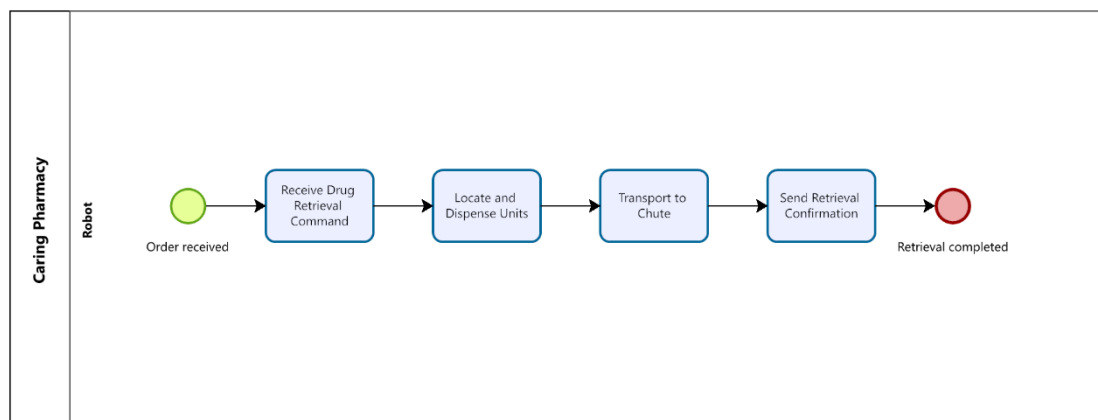


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Figure 10: Level 2 Self Check-in To-Be Diagram

Sub-Process: "Retrieve Drug" (Robot)

This process runs entirely in the **System/Robot** Lane, implementing the **Automation** and **Decomposition heuristics**. By decomposing the original "Fulfill Order" task, we isolated the physical retrieval, allowing the Automated Dispensing System (ADS) to complete the work in approximately 1 minute without human intervention.



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Figure 11: Level 2 Retrieve Drug To-Be Diagram

Sub-Process: "Clinical Safety Check" (Parallel Review)

This diagram represents the specialized path of the parallel fulfillment phase, utilizing **Parallelism** and **Specialization heuristics**. Crucially, this process runs **simultaneously** with the Robot retrieval. Instead of waiting for the drug to be found, the Pharmacist performs the safety check in parallel. This overlapping of tasks is the primary driver of the cycle time reduction.

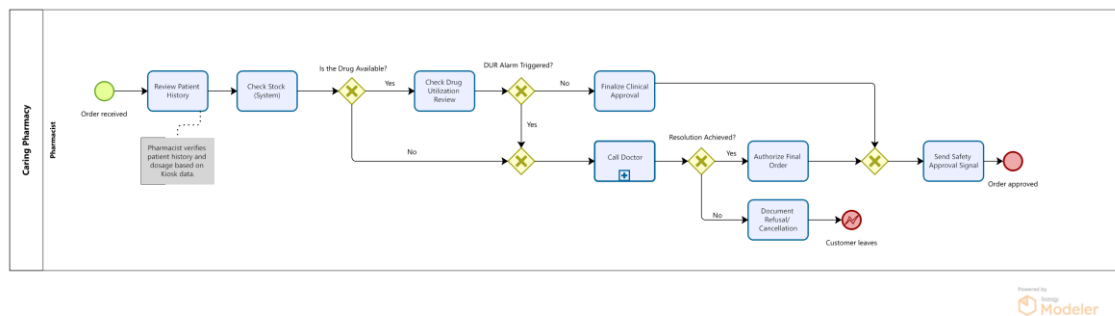


Figure 12: Level 2 Clinical Safety Check To-Be Diagram

Sub-Process: "Enter & Check Details" (Streamlined Eligibility)

Reflecting the **Elimination heuristic**, this diagram shows the simplified eligibility check for prescription customers. The manual data entry tasks previously performed by the Technician have been eliminated. The process now consists of rapid, system-assisted verification steps performed by the Pharmacist, leveraging the pre-validated data captured by the Kiosk.

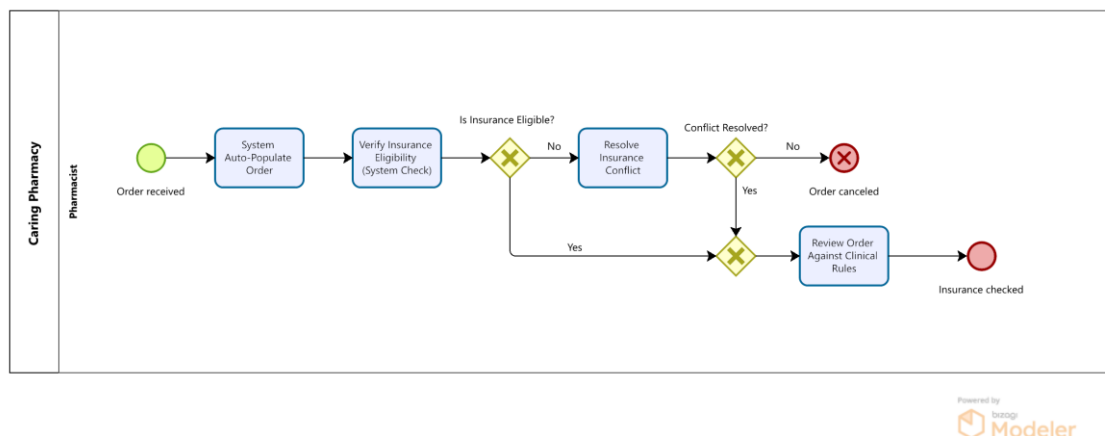


Figure 13: Level 2 Enter & Check Details To-Be Diagram

Sub-Process: "Package & Label Order" (Fast Handover)

This final diagram reflects the simplified endpoint of the process, applying **Elimination** and **Decomposition**. Because payment was eliminated (handled upstream at the Kiosk) and retrieval was decomposed (Robot), the Technician's role is reduced to a simple, linear sequence: Label → Package → Handover. This prevents the Technician from being detained at the counter for administrative tasks.

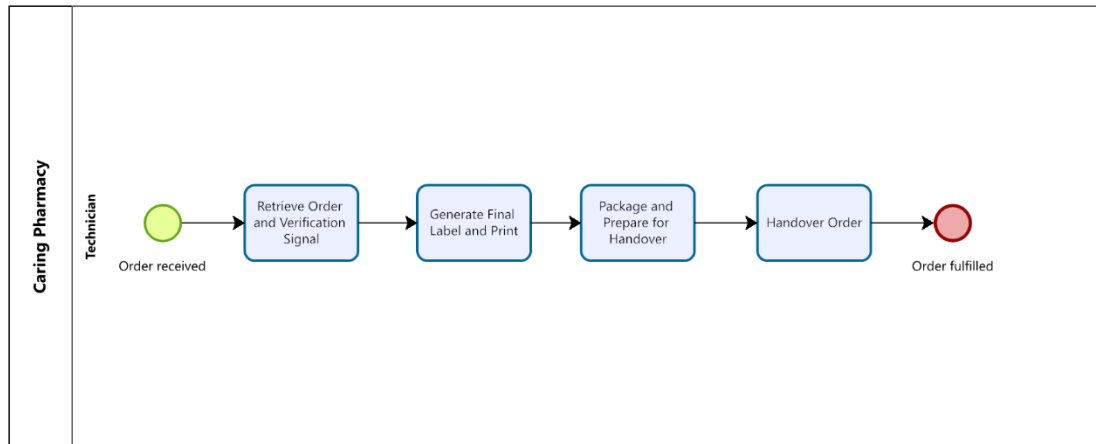
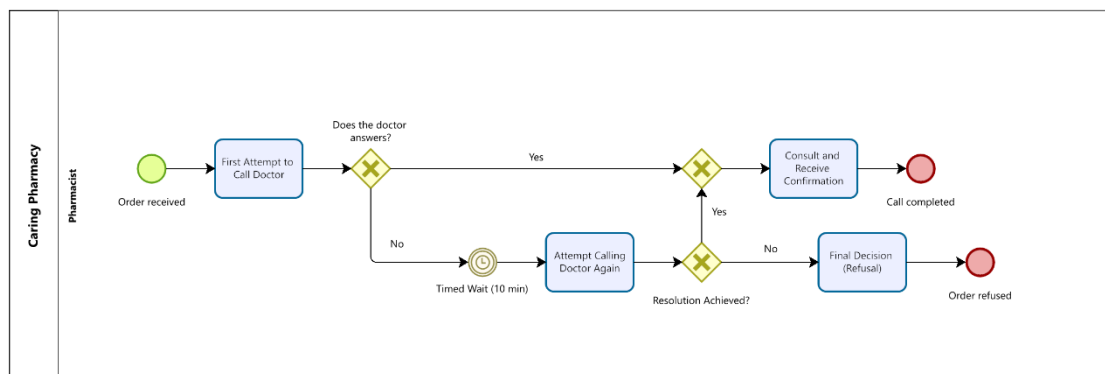


Figure 14: Level 2 Package & Label Order To-Be Diagram

Level 3 Sub-Process: "Call Doctor" (Exception Handling)

Finally, this Level 3 diagram illustrates the robust handling of clinical exceptions. To adhere to the **Elimination of Waste heuristic** (specifically waiting time), we structured a controlled loop with a strictly defined **10-minute timer**. This ensures that if a doctor is unreachable, the process does not hang indefinitely, instead, it forces a resolution (either success or documented refusal) within a predictable timeframe, preserving the overall system flow.



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Figure 15: Level 3 Call Doctor To-Be Diagram

TO-BE Simulation Analysis

Following the structural modeling of the To-Be process, we performed a final simulation to validate the performance of the redesigned workflow. This simulation incorporates the physical "System Lanes" (Kiosk and Robot) and the "Green Lane" logic defined in the BPMN diagrams.

Time and Throughput Results: The structural redesign provided performance metrics that exceeded the initial What-If predictions.

- **Average Cycle Time: 5 minutes and 31 seconds.**
- **Waiting Time: Effectively zero.** The process now flows without interruption, limited only by the physical speed of the machines and the clinical review.

Resource Utilization (The Structural Impact): The explicit assignment of tasks to "System" lanes in the BPMN model revealed the true efficiency of the Technician's new role.

RESOURCE	UTILIZATION	STATUS	ANALYSIS
Technician	12.94%	Free Capacity	By structurally moving "Data Entry" and "Drug Retrieval" to System lanes, the Technician's workload dropped further than predicted (from 22% in What-If to ~13%). They are now purely supervisors and logistical handlers.
Pharmacist	53.10%	Balanced	The Pharmacist remains the steady key point of the process, ensuring clinical safety without becoming a bottleneck.

Table 18: Resource Utilization (The Structural Impact)

Conclusion on Design Validity

The simulation confirms that the **To-Be BPMN model is robust**. The drastic reduction in Technician utilization (to ~13%) confirms that the "System" resources are successfully absorbing the transactional workload, validating the feasibility of the automation strategy.

Sensitivity Analysis: Optimal Staffing Levels

To validate the robustness of the To-Be model and determine the minimum viable staffing level, we performed a sensitivity analysis by reducing the number of Technicians while maintaining the new automated process.

TECHNICIANS	AVG. CYCLE TIME	TECHNICIAN UTIL.	PHARMACIST UTIL.	CONCLUSION
4 (Baseline)	5 min 31 sec	12.94%	53.10%	Highly Scalable: Massive capacity for growth.
3	5 min 31 sec	17.26%	53.10%	Efficient: No impact on customer wait time.
2 (Min.)	5 min 32 sec	25.88%	53.10%	Lean: Still operates with zero queues and healthy slack.

Table 19: Staffing Sensitivity Results

Analysis of Results:

- **Stability:** The cycle time remains virtually constant (~5.5 minutes) regardless of whether we employ 2, 3, or 4 Technicians. This confirms that is the **Process Structure (Automation)** who is now doing the heavy lifting, not the human effort.
- **Minimum Viable Team:** The pharmacy could theoretically operate with just **2 Technicians** (26% utilization) without degrading service speed.
- **Strategic Choice:** While cost-cutting is possible (firing 2 staff), we maintain our recommendation to keep **4 Technicians**. The low utilization (13%) is a strategic asset that allows the pharmacy to handle extreme demand spikes (e.g., pandemics, holidays) that would overwhelm a leaner team.

Comparison of AS-IS vs TO-BE

The implementation of the To-Be model provides transformative improvements in all critical performance metrics. The table below summarizes the shift from the constrained "As-Is" state to the optimized "To-Be" operation.

METRIC	AS-IS (CURRENT)	TO-BE (OPTIMIZED)	PERFORMANCE GAIN
Avg. Cycle Time	12 min 00 sec	5 min 31 sec	54% Faster
Technician Util.	73.76%	12.94%	Load reduced by ~5.7x
Pharmacist Util.	58.87%	53.10%	Ideally Balanced
Throughput Capacity	~230 / day (Max)	~800+ / day (Est)	Capacity Quadrupled

Table20: Performance Comparison

Operational Analysis & Staffing Strategy

Elimination of Waiting Times: The reduction of cycle time to 5 minutes and 31 seconds indicates that the process now operates at the speed of the value-added work itself. The "Inefficiency Gap" (queuing time), which previously accounted for 17% of the customers' journey, has been effectively eliminated by resolving the structural bottlenecks.

Strategic Resource Utilization: Although the simulation indicates the Technician utilization drops to 12.94%, we strongly recommend **retaining the current staff of four Technicians**. This strategic decision prioritizes scalability and resilience over immediate cost-cutting. The low utilization is not waste, instead it represents massive surge capacity, allowing the pharmacy to handle a **300-400% increase in daily volume** without degrading service quality. Also, maintaining the current headcount safeguards the operation against absenteeism and allows for the reinvestment of labor into high-value tasks, such as patient consulting and inventory management, that were previously neglected due to the high operational workload. The 12% utilization is not 'idle time', it is **Surge Capacity**. It effectively insurances the pharmacy against the chaotic demand spikes typical of the healthcare sector (e.g., flu season, pandemics) which the previous AS-IS model could not handle.

Conclusion: The To-Be model successfully resolves the bottleneck, creating a robust, future-proof system that maximizes the ROI of the automation hardware through increased capacity rather than staff reduction.

Financial Analysis

To evaluate the economic feasibility of the To-Be model, we performed a cost-benefit analysis comparing the required initial investment against the projected financial gains.

Implementation Costs (Investment)

The redesign requires a significant upfront Capital Expenditure (CAPEX) to acquire the automation hardware and integrate it with the pharmacy's existing ERP.

ITEM	UNIT COST	QUANTITY	TOTAL COST (€)	NOTES
Automated Dispensing System	€ 85,000	1	€ 85,000	Mid-sized robot unit including installation.
Self-Service Kiosk	€ 6,500	1	€ 6,500	Touchscreen terminal with scanner/reader.
System Integration & Training	€ 5,000	1	€ 5,000	Software linkage and staff training.
Total Initial Investment			€ 96,500	

Table 21: Initial Investment Costs

(Note: We estimate an annual maintenance Operational Expenditure (OPEX) of roughly 10% of the hardware value, or **€ 9,150 / year**).

Strategic Staffing Decision

Although the simulation indicates the Technician utilization drops to 22%, we have made the strategic decision to **retain the current headcount (4 Technicians)**. This decision is driven by a strong financial rationale: the revenue capacity created by the Robot/Technician team allows the pharmacy to process an additional **€198 000** in sales per year, which far outweighs the €14 000 cost of a single technician. Additionally, reducing staff to the theoretical minimum of two creates a fragile system where a single absence could halve throughput and cause customer churn. By maintaining the current team, we ensure operational resilience and avoid the future cost of hiring new staff to handle the projected growth.

Analysis of Benefits (ROI Calculation)

The financial benefits are derived primarily from **Revenue Growth Potential** and **Avoided Costs**.

Current Revenue Calculation

Based on the AS-IS simulation inputs and market averages:

Current Revenue = Daily Orders x Avg. Ticket Price x Operating Days

Current Revenue = 220 Orders/Day x 30€/ Order x 300 Days/ Year =

€1.98M annual revenue

Revenue Growth: We estimate a **10%* increase in daily orders** due to the improved service level and elimination of waiting times. Based on current revenue (€1.98M), this growth projects to **€2 178 000** annually. Assuming a 20% net margin on this extra revenue, the pharmacy gains approximately **€39 600 per year** in net profit.

Avoided Operational Costs: To achieve the optimized 6-minute cycle time using manual labor alone, the simulation indicates we would have needed to hire at least two additional Technicians. By implementing automation instead, we avoid this recurring payroll expense, saving approximately **€28 000 per year** (€14 000 x 2 Technicians).

**The projected 10% revenue growth is a conservative estimate based on the elimination of the 13-minute wait time. In retail environments, reducing queue times is directly correlated with reduced customer churn (abandonment) and increased walk-in traffic. Even if growth is only 5%, the project remains profitable with a Payback Period of roughly 3 years.*

Calculation of Total Annual Benefit

The Total Annual Benefit is the sum of these two streams of financial value:

Total Annual Benefit = Net Profit from Growth + Avoided Operational Costs

Total Annual Benefit = 39 600€ + 28 000€ = **€67 600**

This **€67,600** is the gross cash flow generated or saved annually by the automated system.

Return on Investment (ROI) & Payback

We calculate the ROI based on the **Total Annual Benefit** (€67 600) minus the Annual Maintenance cost (€9 150), resulting in a **Net Annual Cash Flow of €58 450**.

$$ROI = \frac{\text{Net Annual Cash Flow}}{\text{Total Investment}} \times 100$$

$$ROI = \frac{58,450}{96,500} \approx \mathbf{60.5\%}$$

Payback Period:

$$\text{Payback Period} = \frac{\text{Total Investment}}{\text{Net Annual Cash Flow}}$$

$$\text{Payback Period} = \frac{96,500}{58,450} \approx \mathbf{1.65 \text{ Years}}$$

Figure 16: Return on Investment (ROI) & Payback Period Calculation

Conclusion on Feasibility

The financial analysis proves the project is highly viable. The pharmacy recovers the **€96 500** investment in less than **1.7 years**. By prioritizing **capacity growth** over staff reduction, the pharmacy secures a scalable infrastructure that generates positive cash flow significantly faster than hiring human staff to achieve the same level of service.

Implementation and Monitoring Plan

To guarantee the successful transition from the "As-Is" state to the optimized "To-Be" model, we have defined a phased implementation strategy and a continuous monitoring framework.

Implementation Strategy

The introduction of automation (Kiosk and Robot) represents a radical shift in the daily work of the staff. To mitigate resistance and ensure operational continuity, we propose a **Phased Roll-out Approach**:

Phase 1: Front-End Digitalization (Weeks 1-4)

Action: Install the Self-Service Kiosk and train Technicians in the "Supervisory" role.

Goal: Acclimate customers to digital check-in while the manual fulfillment process remains in place. Technicians will actively assist customers to build confidence in the new system.

Phase 2: Back-End Automation (Weeks 5-12)

Action: Install the Automated Dispensing System (Robot) and integrate it with the pharmacy ERP.

Goal: Run the robot in parallel with manual picking for one week to calibrate accuracy. Pharmacists receive training on the new "Parallel Clinical Review" workflow.

Phase 3: Full "Green Lane" Activation (Week 13)

Action: Activate the Triage Gateway logic.

Goal: Non-prescription orders are officially routed to bypass the Pharmacist. The process is now fully live.

Process Monitoring (KPIs)

Once the To-Be process is live, continuous monitoring is required to verify if the real-world performance matches the simulation predictions. We have defined Key Performance Indicators (KPIs) to track success:

KPI	TARGET VALUE (TO-BE)	MEASUREMENT METHOD
Average Cycle Time	< 6 minutes	System timestamp from Kiosk Check-in to Receipt Printing.
Technician Utilization	< 25%	Tracked via time-on-task logs in the ERP. (Ensures surge capacity remains available).
Kiosk Adoption Rate	> 90%	Percentage of customers successfully completing check-in without Technician intervention.
Robot Picking Accuracy	99.9%	Manual audit of dispensed items vs. order manifest.

Table 22: Process Monitoring KPIs

Continuous Improvement: Data from these KPIs will be reviewed monthly. If the **Kiosk Adoption Rate** falls below 90%, additional UI improvements will be implemented to assist customers, ensuring the Technician bottleneck does not re-emerge.

Conclusion

This project successfully identified the operational inefficiencies of Caring Pharmacy and developed a comprehensive redesign strategy that transforms the business from a resource-constrained operation into a highly scalable, technology-driven enterprise.

Important Findings: Through rigorous **Hierarchical Simulation**, we identified that the "As-Is" process was functionally brittle. The centralization of tasks on the **Technician** (who handled 73.76% of the workload) created a structural bottleneck, forcing customers to wait an average of **2 minutes** per visit (17% of their total time). The process logic was also flawed, forcing simple non-prescription customers through unnecessary clinical loops.

Transformative Results: The implementation of the **To-Be model** (Scenario 3) provided results that exceeded initial performance targets. By applying heuristics of **Automation** (Kiosk/Robot), **Triage** (Green Lane), and **Parallelism**, we achieved:

- A **54% reduction** in Average Cycle Time (from 12m 00s to 5m 31s).
- The complete elimination of customer waiting times.
- A reduction in the Technician utilization, from nearly 74% to **12.94%**, effectively breaking the bottleneck.

Strategic & Financial Viability: Financially, the project is highly attractive, offering a **60.5% ROI** and a Payback Period of **1.65 years**. However, the most significant conclusion is strategic: the redesign unlocks massive **surge capacity**. By retaining the current staff despite the efficiency gains, the pharmacy secures the ability to handle a **300-400% increase in volume**.

In summary, the proposed redesign does not only "fix" the current waiting times, it future-proofs the pharmacy, providing the infrastructure required for aggressive revenue growth and long-term operational resilience.

Limitations and Future Work

While the simulation results are robust, the validity of the conclusions is subject to specific limitations inherent in the modeling process.

Simulation Assumptions: The model assumes a steady-state operation with a Poisson arrival rate. In reality, retail environments experience extreme "bursts" (e.g., lunch hour rushes) that might temporarily stress the Kiosk capacity more than the average indicates. Consequently, the model assumes **99% uptime** for the Robot and Kiosk, thus real-world mechanical failures would require fallback protocols not modeled here.

Human Factors and Adoption: The model assumes a **95% adoption rate** for the Self-Service Kiosk. If the local demographic is resistant to technology (e.g., elderly population), the "Resolve Exception" path would trigger more frequently than the modeled 5%, potentially re-introducing a minor bottleneck at the Technician level. Future work should include a "Change Management" study to assess customer digital literacy.

Scope of Analysis: The financial analysis focuses on the "Order-to-Cash" process. It does not account for potential inventory savings (reduced shrinkage/waste) provided by the Robot, nor does it account for the potential increase in utility costs (electricity for automation). A holistic P&L analysis would refine the ROI calculation further.

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Appendix

Appendix A: To-Be Simulation Visualizations

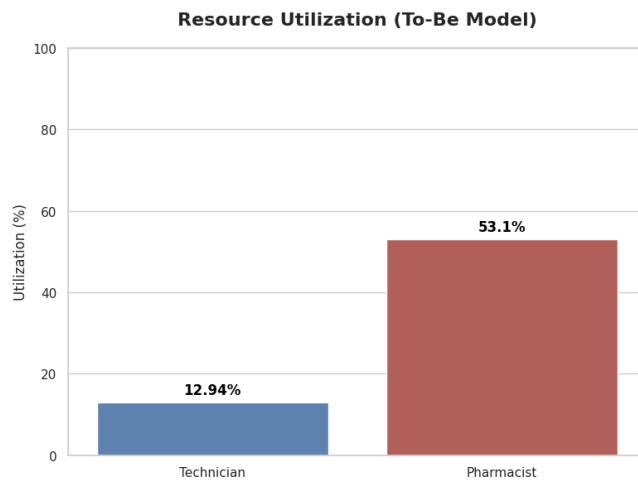


Figure A 1: Resource Utilization Profile (To-Be Model)

This chart illustrates the redistribution of workload in the optimized model. The **Technician utilization (12.94%)** reflects the successful offloading of transactional tasks to automation, creating significant surge capacity. The **Pharmacist utilization (53.10%)** remains balanced, ensuring clinical safety tasks are performed without creating a new bottleneck.

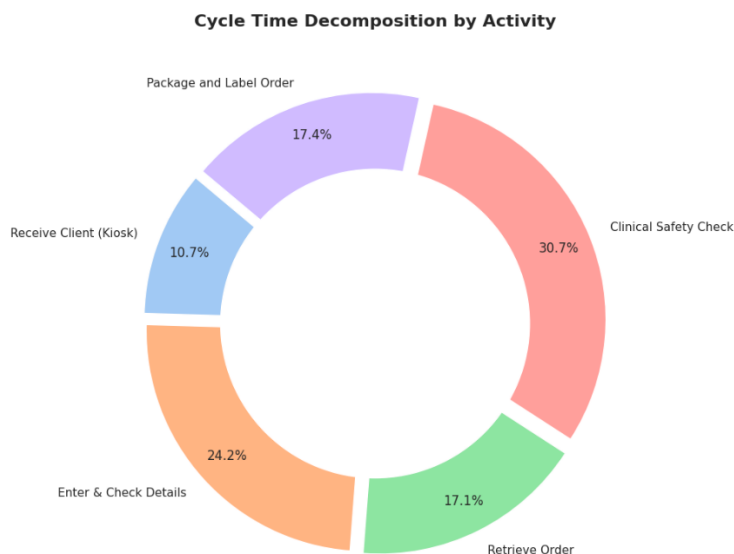


Figure A 2: Cycle Time Decomposition

A breakdown of the total cycle time (5.5 minutes) by activity type. Most of the process time is now dedicated to high-value **Clinical Safety Checks** and **Enter & Check Details** (Pharmacist tasks), while logistical steps like **Receive Client** and **Retrieve Order** have been minimized through Kiosk and Robot integration.

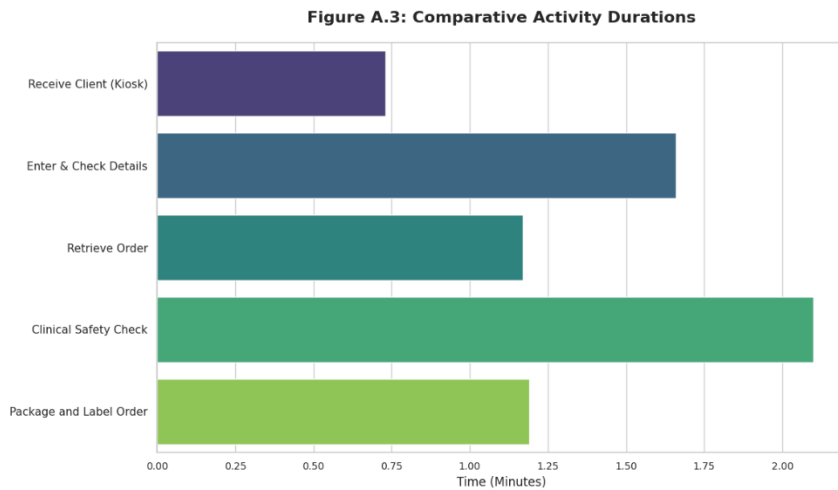


Figure A 3: Comparative Activity Durations

*This comparison highlights the efficiency gains from automation. The **Retrieve Order** task (Robot) and **Receive Client** task (Kiosk) operate at high speeds (approx. 1 minute or less), significantly faster than the manual equivalents in the As-Is model. The **Clinical Safety Check** remains the longest task, appropriately reflecting the human cognitive load required for patient safety.*

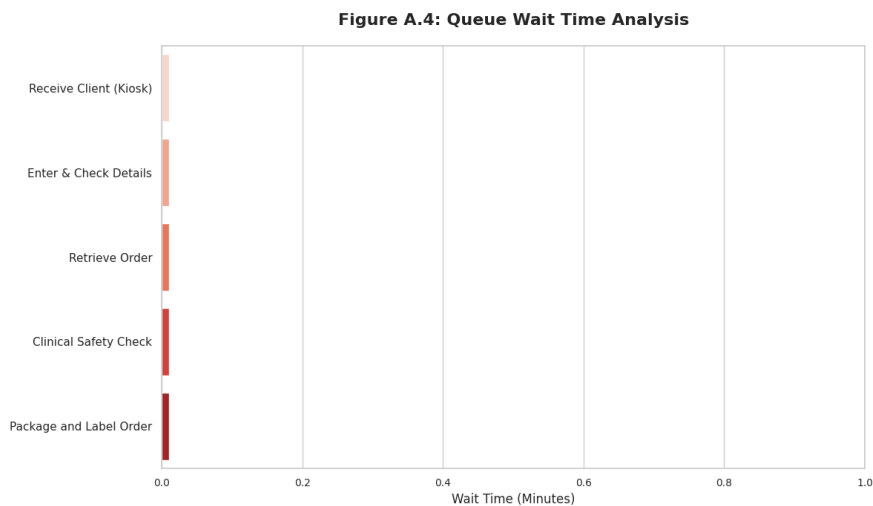


Figure A 4: Queue Wait Time Analysis

*Analysis of average waiting times per activity in the To-Be scenario. The chart demonstrates that wait times have been effectively reduced to **near-zero** (<1 second) across all activities. This confirms that the structural changes (Parallelism and Triage) have successfully eliminated the resource contention that caused the 3.5-minute delay in the As-Is state.*

Appendix B: Financial Projections

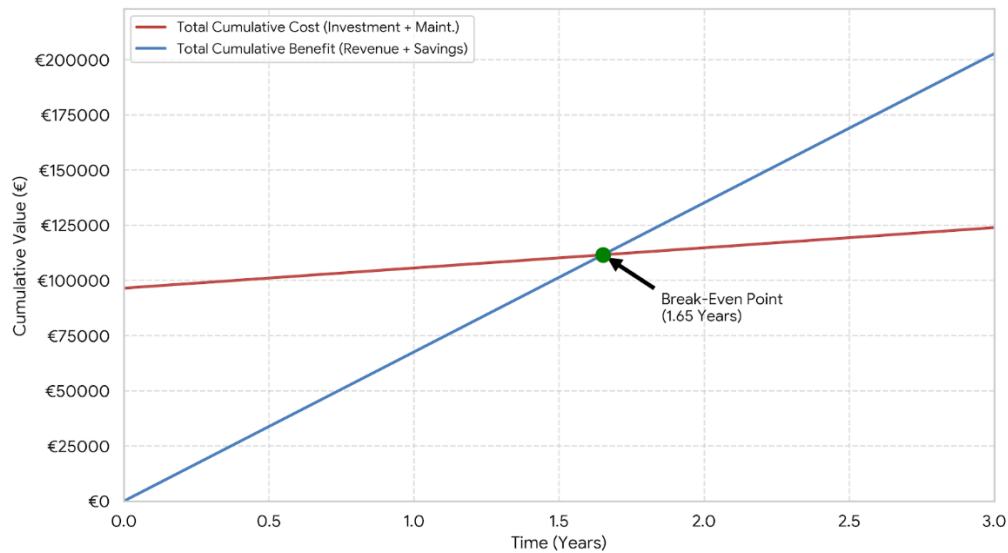


Figure B 1: Cumulative Cash Flow & Break-Even Analysis

A projection of financial performance over the first three years post-implementation. The red line represents the **Total Cumulative Cost**, comprising the initial €96 500 investment and recurring maintenance expenses. The blue line tracks the **Total Cumulative Benefit**, derived from projected revenue growth and avoided operational costs. The intersection point at approximately **1.65 years** visually confirms the project's Payback Period, after which the automation investment generates continuous positive cash flow for the pharmacy.

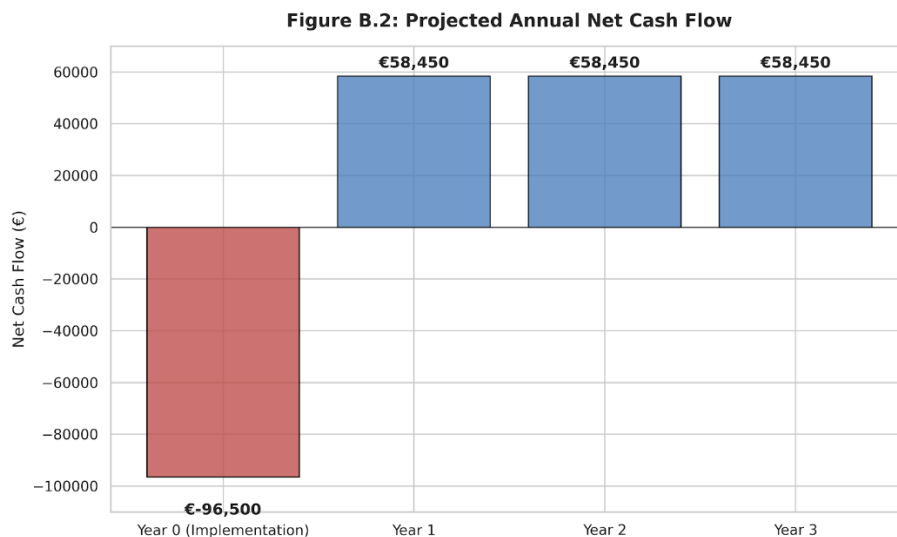


Figure B 2: Projected Annual Net Cash Flow

A visualization of the cash flow dynamics. The significant initial outflow in Year 0 (€96 500) represents the capital expenditure for the Robot and Kiosk. In subsequent years, the project generates a consistent positive net cash flow of €58 450 annually, derived from operational savings and revenue growth, ensuring financial sustainability.

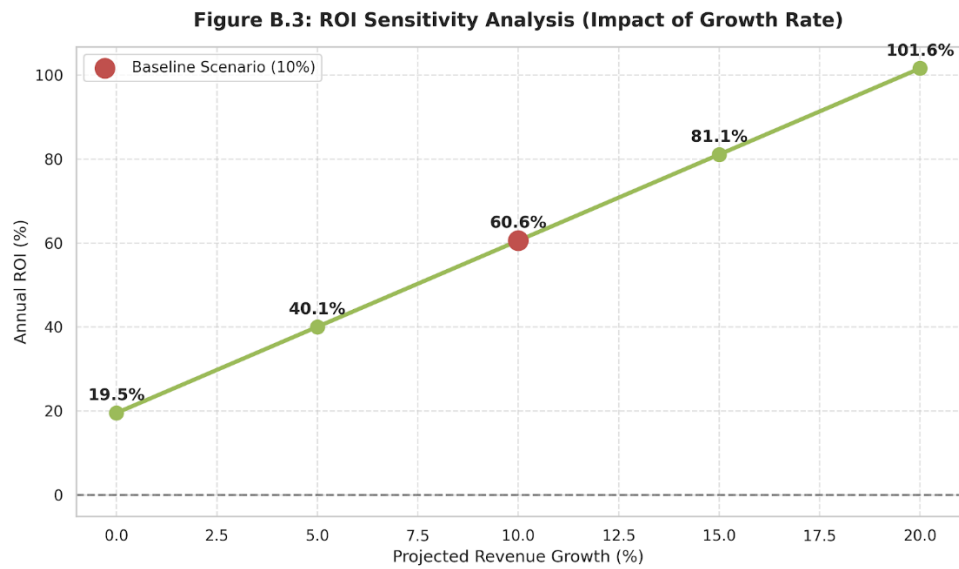


Figure B 3: ROI Sensitivity Analysis (Risk Assessment)

*A stress test of the financial model against varying revenue growth scenarios. The chart demonstrates that even in a worst-case scenario with **0% revenue growth** (relying solely on avoided operational costs), the project still yields a positive ROI of approx. 19.5%. This confirms that the investment is financially secure and not solely dependent on optimistic market expansion.*

Appendix C: Implementation and Impact

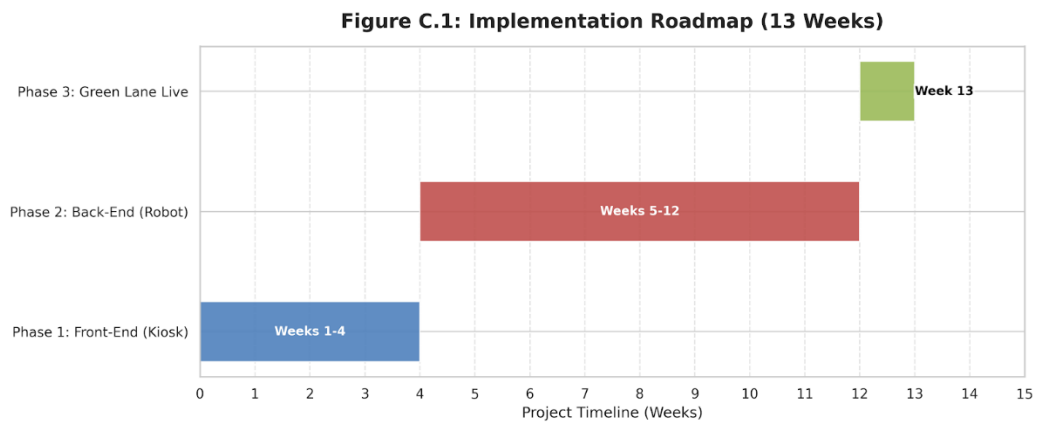


Figure C 1: Implementation Roadmap

A phased timeline for the roll-out of the To-Be model. The schedule prioritizes "Front-End Digitalization" (Weeks 1-4) to acclimate customers to the Kiosk before introducing the more complex "Back-End Automation" (Robot). This staggered approach mitigates operational risk.

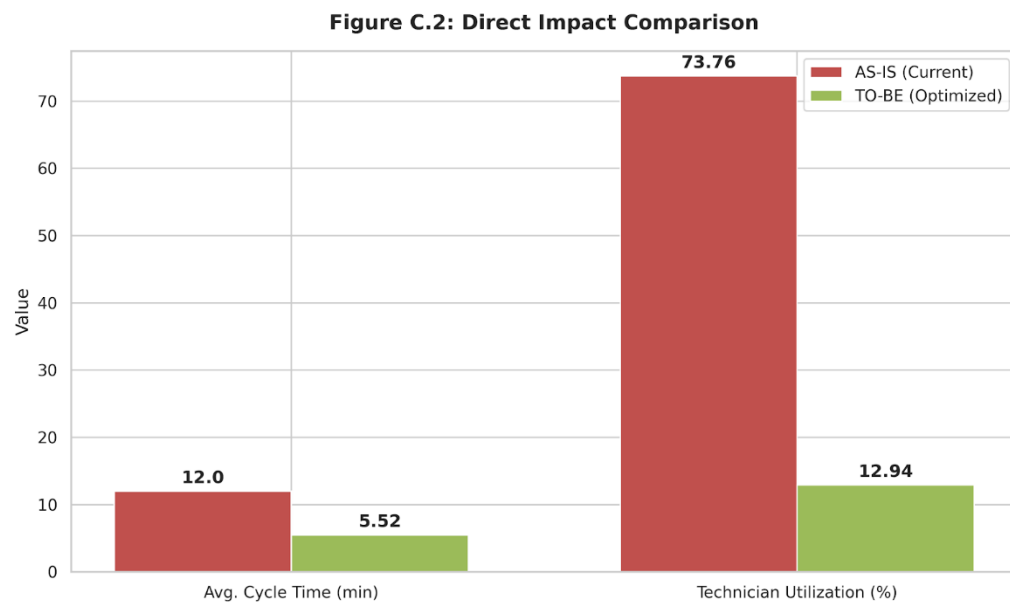


Figure C 2: Direct Impact Comparison (AS-IS vs TO-BE)

A side-by-side comparison of the critical Key Performance Indicators (KPIs). The chart visually demonstrates the magnitude of the improvement: Cycle Time is halved, and Technician Utilization is slashed by nearly 6x, validating the effectiveness of the redesign.